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Benjamin Georgette, David Dureisseix, Anthony Gravouil. Derivation of the Jacobi identity for continuum thermodynamics: application to thermo-visco-elastodynamics. INSA Lyon, CNRS, LaMCoS, UMR5259, 69621 Villeurbanne, France. 2025. <hal-05223598>

HAL Id: hal-05223598

<https://hal.science/hal-05223598v1>

Submitted on 26 Aug 2025

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Derivation of the Jacobi identity for continuum thermodynamics: application to thermo-visco-elastodynamics

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August 26, 2025

The Double Bracket Structure is a variational bracket framework. This *research report* presents various details of calculus that have been required to derive the results of a main paper, submitted to a peer-reviewed journal. First, the structure is derived for a uni-dimensional small strain generalized standard material, with a general quadratic dissipation potential. Second, it is developed for the example of large strain thermo-visco-elastodynamics, within the multisymplectic framework.

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Nomenclature

General notations

$\mathcal{F}$	General functional of density f
$\hat{e}_{\text{int}}$	Density of internal energy
$\hat{\mathcal{E}}_{\text{tot}}$	Total energy functional, of density $\hat{E}_{\text{tot}}$
$\mathcal{L}_v$	Lie derivative
ω_X	Spatial subdomain of the Lagrangian configuration
ρ	Density
$\mathcal{S}$	Total entropy functional, of density s
c	Thermal capacity
$K, \underline{\underline{K}}$	Thermal conductivity (scalar coefficient, tensor)
$p, \underline{p}$	Linear momentum (scalar, tensorial)
$q, \underline{q}$	Thermal conduction (scalar, tensorial)
T	Absolute temperature
w	Helmholtz free energy
Ω_X	Lagrangian configuration of the continuum

Notations specific to the large strain, tri-dimensional problem

$\underline{\chi}$	Configuration mapping
$\underline{\underline{C}}$	Cauchy-Green dilatation tensor
$\underline{\underline{E}}$	Euler-Lagrange strain tensor
$\underline{\tau}$	Kirchhoff stress tensor
$\underline{\underline{S}}$	Second Piola-Kirchhoff stress tensor
$\underline{\underline{\Pi}}$	First Piola-Kirchhoff stress tensor
$\underline{\underline{F}}$	Gradient of transformation
$\underline{\underline{F}}_e$	Elastic part of the gradient of transformation
$\underline{\underline{F}}_i$	Inelastic part of the gradient of transformation
$\underline{\underline{N}}$	Lagrangian viscous flow tensor
$\underline{\underline{V}}$	Eulerian viscous flow tensor

w_{NEQ} Out of equilibrium Helmholtz free energy

Notations specific to the small strain uni-dimensional problem

V	Matrix of dissipation coefficients
ϕ	Dissipation potential
σ	Cauchy stress tensor
$\alpha = (\alpha_1, \dots, \alpha_N)$	N internal variables
ε	(Scalar) small strain linearized tensor
u	(Scalar) Displacement

1. Uni-dimensional small strain generalized standard material

The Double Bracket Structure for a uni-dimensional generalized standard material at small strain is shown to be equal to

$$\frac{d\mathcal{F}}{dt} = \{\mathcal{F}, \hat{\mathcal{E}}_{\text{tot}}\} + (\mathcal{F}, \mathcal{S}) + \quad (1)$$

$$+ \int_{\partial\omega_X} \frac{\partial f}{\partial \varepsilon} \frac{\partial u}{\partial t} d(\partial\omega_X) - \int_{\partial\omega_X} \frac{\partial f}{\partial p} \left(\frac{\partial s}{\partial T} \right)^{-1} \frac{\partial s}{\partial \varepsilon} \frac{\partial \hat{E}_{\text{tot}}}{\partial T} d(\partial\omega_X) - \int_{\partial\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} K \frac{T^2}{c^2} \frac{\partial}{\partial X} \left(\frac{\partial s}{\partial T} \right) d(\partial\omega_X) \quad (2)$$

where

$$\begin{aligned} \{\mathcal{F}, \hat{\mathcal{E}}_{\text{tot}}\} = & \int_{\omega_X} \left[\left(\frac{\partial f}{\partial u} - \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial \varepsilon} \right) \right) \frac{\partial \hat{E}_{\text{tot}}}{\partial p} - \frac{\partial f}{\partial p} \left(\frac{\partial \hat{E}_{\text{tot}}}{\partial u} - \frac{\partial}{\partial X} \left(\frac{\partial \hat{E}_{\text{tot}}}{\partial \varepsilon} \right) \right) + \right. \\ & \left. + \left(\frac{\partial s}{\partial T} \right)^{-1} \frac{\partial s}{\partial \varepsilon} \left(\frac{\partial \hat{E}_{\text{tot}}}{\partial T} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial p} \right) - \frac{\partial f}{\partial T} \frac{\partial}{\partial X} \left(\frac{\partial \hat{E}_{\text{tot}}}{\partial p} \right) \right) \right] d\omega_X \end{aligned} \quad (3)$$

$$\begin{aligned} (\mathcal{F}, \mathcal{S}) = & \int_{\omega_X} \left[\frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) K \frac{T^2}{c^2} \frac{\partial}{\partial X} \left(\frac{\partial s}{\partial T} \right) + \frac{\partial f}{\partial T} \frac{1}{c} \frac{T}{c} \left(\frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \right)^T \mathbf{V}^{-1} \frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \frac{\partial s}{\partial T} + \right. \\ & \left. - \frac{\partial f}{\partial T} \frac{T}{c} \left(\frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \right)^T \mathbf{V}^{-1} \frac{\partial s}{\partial \alpha} - \frac{T}{c} \left(\frac{\partial f}{\partial \alpha} \right)^T \mathbf{V}^{-1} \frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \frac{\partial s}{\partial T} + T \left(\frac{\partial f}{\partial \alpha} \right)^T \mathbf{V}^{-1} \frac{\partial s}{\partial \alpha} \right] d\omega_X \end{aligned} \quad (4)$$

1.1. Evolution of temperature

Proposition 1.1

The time derivative of the temperature is given by

$$\frac{\partial T}{\partial t} = -\frac{1}{\rho c} \frac{\partial q}{\partial X} - \frac{1}{c} \left(\frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \right)^T \frac{\partial \alpha}{\partial t} - \frac{T}{c} \frac{\partial s}{\partial \varepsilon} \frac{\partial \varepsilon}{\partial t} \quad (5)$$

Proof _____

Consider the time derivative of the Gibbs equation [truesdell-1984a]

$$\frac{\partial}{\partial t} (\rho \hat{e}_{\text{int}}) = \frac{\partial}{\partial t} (\rho w + \rho Ts) \quad (6)$$

$$\Leftrightarrow \rho \dot{\hat{e}}_{\text{int}} = \rho \dot{w} - \rho \dot{T} \frac{\partial w}{\partial T} - \rho T \frac{\partial}{\partial t} \frac{\partial w}{\partial T} \quad (7)$$

$$\Leftrightarrow \sigma \frac{\partial \varepsilon}{\partial t} - \frac{\partial q}{\partial X} = \rho \frac{\partial w}{\partial \varepsilon} + \rho \frac{\partial w}{\partial T} \dot{T} + \rho \sum_{i=1}^N \frac{\partial w}{\partial \alpha_i} \frac{\partial \alpha_i}{\partial t} - \rho \dot{T} \frac{\partial w}{\partial T} + \quad (8)$$

$$- \rho T \frac{\partial^2 w}{\partial T \partial \varepsilon} \frac{\partial \varepsilon}{\partial t} - \rho T \frac{\partial^2 w}{\partial T^2} \dot{T} - \rho T \sum_{i=1}^N \frac{\partial^2 w}{\partial T \partial \alpha_i} \frac{\partial \alpha_i}{\partial t} \quad (9)$$

$$\Leftrightarrow \left(\sigma - \rho \frac{\partial w}{\partial \varepsilon} \right) \frac{\partial \varepsilon}{\partial t} - \frac{\partial q}{\partial X} + \rho T \frac{\partial^2 w}{\partial T \partial \varepsilon} \frac{\partial \varepsilon}{\partial t} = - \rho T \underbrace{\frac{\partial^2 w}{\partial T^2}}_{:= \rho c} \dot{T} + \rho \sum_{i=1}^N \frac{\partial}{\partial \alpha_i} (w + Ts) \frac{\partial \alpha_i}{\partial t} \quad (10)$$

$$\Leftrightarrow -\frac{\partial q}{\partial X} + \rho T \frac{\partial^2 w}{\partial T \partial \varepsilon} \frac{\partial \varepsilon}{\partial t} - \rho \sum_{i=1}^N \frac{\partial \hat{e}_{\text{int}}}{\partial \alpha_i} \dot{\alpha}_i = \rho c \dot{T} \quad (11)$$

$$\Leftrightarrow \frac{\partial T}{\partial t} = -\frac{1}{\rho c} \frac{\partial q}{\partial X} - \frac{1}{c} \sum_{i=1}^N \frac{\partial \hat{e}_{\text{int}}}{\partial \alpha_i} \frac{\partial \alpha_i}{\partial t} - \frac{T}{c} \frac{\partial s}{\partial \varepsilon} \frac{\partial \varepsilon}{\partial t} \quad (12)$$

$$\Leftrightarrow \frac{\partial T}{\partial t} = -\frac{1}{\rho c} \frac{\partial q}{\partial X} - \frac{1}{c} \left(\frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \right)^T \frac{\partial \alpha}{\partial t} - \frac{T}{c} \frac{\partial s}{\partial \varepsilon} \frac{\partial \varepsilon}{\partial t} \quad (13)$$

_____ *qed.* ■

1.2. Integral by parts of the conduction part of the dissipative bracket

Proposition 1.2

The integral within the Double Bracket Structure concerning the conduction evolution of temperature yields

$$\begin{aligned} & \int_{\omega_X} \frac{\partial f}{\partial T} \frac{1}{\rho c} \frac{\partial}{\partial X} \left(K \frac{\partial T}{\partial X} \right) d\omega_X \\ &= \int_{\omega_X} \frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) K \frac{T^2}{c^2} \frac{\partial}{\partial X} \left(\frac{\partial s}{\partial T} \right) d\omega_X - \int_{\partial\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} K \frac{T^2}{c^2} \frac{\partial}{\partial X} \left(\frac{\partial s}{\partial T} \right) d(\partial\omega_X) \end{aligned} \quad (14)$$

Proof _____

First, notice that [Lemaitre and Chaboche 2001]

$$c = T \frac{\partial s}{\partial T} \quad (15)$$

Then, one obtains

$$\int_{\omega_X} \frac{\partial f}{\partial T} \frac{1}{\rho c} \frac{\partial}{\partial X} \left(K \frac{\partial T}{\partial X} \right) d\omega_X = \int_{\omega_X} \frac{\partial f}{\partial T} \frac{1}{\rho c} \frac{\partial}{\partial X} \left(K \frac{\partial}{\partial X} \left(\frac{c}{\frac{\partial s}{\partial T}} \right) \right) d\omega_X \quad (16)$$

$$= - \int_{\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} \frac{1}{c} \frac{\partial}{\partial X} \left(K c \frac{1}{\left(\frac{\partial s}{\partial T} \right)^2} \frac{\partial}{\partial X} \left(\frac{\partial s}{\partial T} \right) \right) d\omega_X \quad (17)$$

$$= - \int_{\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} \frac{1}{c} \frac{\partial}{\partial X} \left(K \frac{1}{c} T^2 \frac{\partial}{\partial X} \left(\frac{\partial s}{\partial T} \right) \right) d\omega_X \quad (18)$$

$$= \int_{\omega_X} \frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) K \frac{T^2}{c^2} \frac{\partial}{\partial X} \left(\frac{\partial s}{\partial T} \right) d\omega_X - \int_{\partial\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} K \frac{T^2}{c^2} \frac{\partial}{\partial X} \left(\frac{\partial s}{\partial T} \right) d(\partial\omega_X) \quad (19)$$

by integration by parts.

_____ *qed.* ■

1.3. Skew-symmetry of the reversible bracket

Proposition 1.3

The reversible bracket, defined in (3) is skew-symmetric.

Proof _____

$$\begin{aligned} & \{\mathcal{F}, \mathcal{G}\} + \{\mathcal{G}, \mathcal{F}\} \\ &= \int_{\omega_X} \left[\underbrace{\left(\frac{\partial f}{\partial u} - \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial \varepsilon} \right) \right) \frac{\partial g}{\partial p}}_{(a)} - \underbrace{\frac{\partial f}{\partial p} \left(\frac{\partial g}{\partial u} - \frac{\partial}{\partial X} \frac{\partial g}{\partial \varepsilon} \right)}_{-(b)} + \underbrace{\left(\frac{\partial s}{\partial T} \right)^{-1} \frac{\partial s}{\partial \varepsilon} \left(\frac{\partial g}{\partial T} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial p} \right) - \frac{\partial f}{\partial T} \frac{\partial}{\partial X} \left(\frac{\partial g}{\partial p} \right) \right)}_{(c)} - \underbrace{\frac{\partial f}{\partial T} \frac{\partial}{\partial X} \left(\frac{\partial g}{\partial p} \right)}_{-(d)} \right] \\ &+ \left[\underbrace{\left(\frac{\partial g}{\partial u} - \frac{\partial}{\partial X} \left(\frac{\partial g}{\partial \varepsilon} \right) \right) \frac{\partial f}{\partial p}}_{(b)} - \underbrace{\frac{\partial g}{\partial p} \left(\frac{\partial f}{\partial u} - \frac{\partial}{\partial X} \frac{\partial f}{\partial \varepsilon} \right)}_{-(a)} + \underbrace{\left(\frac{\partial s}{\partial T} \right)^{-1} \frac{\partial s}{\partial \varepsilon} \left(\frac{\partial f}{\partial T} \frac{\partial}{\partial X} \left(\frac{\partial g}{\partial p} \right) - \frac{\partial g}{\partial T} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial p} \right) \right)}_{(d)} - \underbrace{\frac{\partial g}{\partial T} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial p} \right)}_{-(c)} \right] d\omega_X \\ &= 0 \end{aligned} \quad (20)$$

_____ *qed.* ■

1.4. The Jacobi identity

Proposition 1.4

The reversible bracket (3) is consistent with the Jacobi identity

$$\{\{\mathcal{F}, \mathcal{G}\}, \mathcal{H}\} + \{\{\mathcal{G}, \mathcal{H}\}, \mathcal{F}\} + \{\{\mathcal{H}, \mathcal{F}\}, \mathcal{G}\} = 0 \quad \forall \mathcal{F}, \mathcal{G}, \mathcal{H} \quad (21)$$

Proof

We will show that the densities of the functionals verify the Jacobi identity, thus proving that it is verified for the functionals (while the other sense might not be true)

$$\{\{f, g\}, h\} + \{\{g, h\}, f\} + \{\{h, f\}, g\} = 0 \quad (22)$$

We consider the following assumptions

$$\frac{\partial s}{\partial p} = 0, \quad \frac{\partial s}{\partial u} = 0, \quad \frac{\partial T}{\partial p} = 0 \quad (23)$$

consistent with the specific expression of the entropy that will be proposed later. The reversible bracket (on densities) is given by

$$\{f, g\} = \underbrace{\left(\frac{\partial f}{\partial u} - \frac{\partial}{\partial X} \frac{\partial f}{\partial \varepsilon} \right) \frac{\partial g}{\partial p} + \frac{\partial f}{\partial p} \left(-\frac{\partial g}{\partial u} + \frac{\partial}{\partial X} \frac{\partial g}{\partial \varepsilon} \right)}_{:=\{f, g\}_{\text{canonical}}} + \underbrace{\left(\frac{\partial s}{\partial T} \right)^{-1} \frac{\partial s}{\partial \varepsilon} \left(\frac{\partial g}{\partial T} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial p} \right) - \frac{\partial f}{\partial T} \frac{\partial}{\partial X} \left(\frac{\partial g}{\partial p} \right) \right)}_{:=\{f, g\}_{\text{thermal}}} \quad (24)$$

Therefore, we have, for one part of the Jacobi identity

$$\{\{f, g\}, h\} = \{\{f, g\}_{\text{canonical}} + \{f, g\}_{\text{thermal}}, h\} = \{\{f, g\}_{\text{canonical}}, h\} + \{\{f, g\}_{\text{thermal}}, h\} \quad (25)$$

$$= \{\{f, g\}_{\text{canonical}}, h\}_{\text{canonical}} + \{\{f, g\}_{\text{canonical}}, h\}_{\text{thermal}} + \quad (26)$$

$$+ \{\{f, g\}_{\text{thermal}}, h\}_{\text{canonical}} + \{\{f, g\}_{\text{thermal}}, h\}_{\text{thermal}} \quad (27)$$

Thus, we will proceed to the verification of the Jacobi identity in the following manner

$$\{\{f, g\}, h\} + \{\{g, h\}, f\} + \{\{h, f\}, g\} \quad (28)$$

$$= \{\{f, g\}_{\text{canonical}}, h\}_{\text{canonical}} + \{\{f, g\}_{\text{canonical}}, h\}_{\text{thermal}} + \quad (29)$$

$$+ \{\{f, g\}_{\text{thermal}}, h\}_{\text{canonical}} + \{\{f, g\}_{\text{thermal}}, h\}_{\text{thermal}} + \quad (30)$$

$$+ \{\{g, h\}_{\text{canonical}}, f\}_{\text{canonical}} + \{\{g, h\}_{\text{canonical}}, f\}_{\text{thermal}} + \quad (31)$$

$$+ \{\{g, h\}_{\text{thermal}}, f\}_{\text{canonical}} + \{\{g, h\}_{\text{thermal}}, f\}_{\text{thermal}} + \quad (32)$$

$$+ \{\{h, f\}_{\text{canonical}}, g\}_{\text{canonical}} + \{\{h, f\}_{\text{canonical}}, g\}_{\text{thermal}} + \quad (33)$$

$$+ \{\{h, f\}_{\text{thermal}}, g\}_{\text{canonical}} + \{\{h, f\}_{\text{thermal}}, g\}_{\text{thermal}} \quad (34)$$

$$= \{\{f, g\}_{\text{canonical}}, h\}_{\text{canonical}} + \{\{g, h\}_{\text{canonical}}, f\}_{\text{canonical}} + \{\{h, f\}_{\text{canonical}}, g\}_{\text{canonical}} + \quad (35)$$

$$+ \{\{f, g\}_{\text{canonical}}, h\}_{\text{thermal}} + \{\{f, g\}_{\text{thermal}}, h\}_{\text{canonical}} + \{\{g, h\}_{\text{canonical}}, f\}_{\text{thermal}} + \quad (36)$$

$$+ \{\{g, h\}_{\text{thermal}}, f\}_{\text{canonical}} + \{\{h, f\}_{\text{canonical}}, g\}_{\text{thermal}} + \{\{h, f\}_{\text{thermal}}, g\}_{\text{canonical}} + \quad (37)$$

$$+ \{\{f, g\}_{\text{thermal}}, h\}_{\text{thermal}} + \{\{g, h\}_{\text{thermal}}, f\}_{\text{thermal}} + \{\{h, f\}_{\text{thermal}}, g\}_{\text{thermal}} \quad (38)$$

We will consider that the functions are continuous in every variable, such that the Schwartz identity is verified.

Canonical - canonical

First, we have

$$\{\{f, g\}_{\text{canonical}}, h\}_{\text{canonical}} + \{\{g, h\}_{\text{canonical}}, f\}_{\text{canonical}} + \{\{h, f\}_{\text{canonical}}, g\}_{\text{canonical}} \quad (39)$$

$$= \left\{ \left(\frac{\partial f}{\partial u} - \frac{\partial}{\partial X} \frac{\partial f}{\partial \varepsilon} \right) \frac{\partial g}{\partial p} + \frac{\partial f}{\partial p} \left(-\frac{\partial g}{\partial u} + \frac{\partial}{\partial X} \frac{\partial g}{\partial \varepsilon} \right), h \right\}_{\text{canonical}} + \quad (40)$$

$$+ \left\{ \left(\frac{\partial g}{\partial u} - \frac{\partial}{\partial X} \frac{\partial g}{\partial \left(\frac{\partial u}{\partial X} \right)} \right) \frac{\partial h}{\partial p} + \frac{\partial g}{\partial p} \left(-\frac{\partial h}{\partial u} + \frac{\partial}{\partial X} \frac{\partial h}{\partial \varepsilon} \right), f \right\}_{\text{canonical}} + \quad (41)$$

$$+ \left\{ \left(\frac{\partial h}{\partial u} - \frac{\partial}{\partial X} \frac{\partial h}{\partial \left(\frac{\partial u}{\partial X} \right)} \right) \frac{\partial f}{\partial p} + \frac{\partial h}{\partial p} \left(-\frac{\partial f}{\partial u} + \frac{\partial}{\partial X} \frac{\partial f}{\partial \varepsilon} \right), g \right\}_{\text{canonical}} \quad (42)$$

$$= \{(f, u - f, \varepsilon X) g, p + f, p (-g, u + g, \varepsilon X), h\}_{\text{canonical}} + \quad (43)$$

$$+ \{(g, u - g, \varepsilon X) h, p + g, p (-h, u + h, \varepsilon X), f\}_{\text{canonical}} + \quad (44)$$

$$+ \{(h, u - h, \varepsilon X) f, p + h, p (-f, u + f, \varepsilon X), g\}_{\text{canonical}} \quad (45)$$

$$= ((f, u - f, \varepsilon X) g, p + f, p (-g, u + g, \varepsilon X))_{,u} - ((f, u - f, \varepsilon X) g, p + f, p (-g, u + g, \varepsilon X))_{,\varepsilon X} h, p + \quad (46)$$

$$+ ((f, u - f, \varepsilon X) g, p + f, p (-g, u + g, \varepsilon X))_{,p} (-h, u + h, \varepsilon X) + \quad (47)$$

$$+ (((g, u - g, \varepsilon X) h, p + g, p (-h, u + h, \varepsilon X))_{,u} - ((g, u - g, \varepsilon X) h, p + g, p (-h, u + h, \varepsilon X))_{,\varepsilon X}) f, p + \quad (48)$$

$$+ ((g, u - g, \varepsilon X) h, p + g, p (-h, u + h, \varepsilon X))_{,p} (-f, u + f, \varepsilon X) + \quad (49)$$

$$+ (((h, u - h, \varepsilon X) f, p + h, p (-f, u + f, \varepsilon X))_{,u} - ((h, u - h, \varepsilon X) f, p + h, p (-f, u + f, \varepsilon X))_{,\varepsilon X}) g, p + \quad (50)$$

$$+ ((h, u - h, \varepsilon X) f, p + h, p (-f, u + f, \varepsilon X))_{,p} (-g, u + g, \varepsilon X) + \quad (51)$$

$$= \underbrace{f, uu g, p h, p}_{=(1)} - \underbrace{f, \varepsilon X u g, p h, p}_{=-(2)} + \underbrace{f, u g, pu h, p}_{=(3)} - \underbrace{f, \varepsilon X g, pu h, p}_{=-(4)} - \underbrace{f, pu g, u h, p}_{=-(5)} + \underbrace{f, pu g, \varepsilon X h, p}_{=(6)} - \underbrace{f, p g, uu h, p}_{=-(7)} \quad (52)$$

$$+ \underbrace{f, p g, \varepsilon X u h, p}_{=(8)} - \underbrace{f, u \varepsilon X g, p h, p}_{=-(9)} + \underbrace{f, \varepsilon X \varepsilon X g, p h, p}_{=(10)} - \underbrace{f, u g, p \varepsilon X h, p}_{=-(11)} + \underbrace{f, \varepsilon X g, p \varepsilon X h, p}_{=(12)} + \underbrace{f, p \varepsilon X g, u h, p}_{=(13)} \quad (53)$$

$$- \underbrace{f, p \varepsilon X g, \varepsilon X h, p}_{=-(14)} + \underbrace{f, p g, u \varepsilon X h, p}_{=(15)} - \underbrace{f, p g, \varepsilon X \varepsilon X h, p}_{=-(16)} - \underbrace{h, u f, up g, p}_{=-(17)} + \underbrace{h, u f, \varepsilon X p g, p}_{=(18)} - \underbrace{h, u f, u g, pp}_{=-(19)} + \quad (54)$$

$$+ \underbrace{h, u f, \varepsilon X g, pp}_{=(20)} + \underbrace{h, u f, pp g, u}_{=(21)} - \underbrace{h, u f, pp g, \varepsilon X}_{=-(22)} + \underbrace{h, u f, p g, up}_{=(23)} - \underbrace{h, u f, p g, \varepsilon X p}_{=-(24)} + \underbrace{h, \varepsilon X f, up g, p}_{=(25)} + \quad (55)$$

$$+ \underbrace{-h, \varepsilon X f, \varepsilon X p g, p}_{=-(26)} + \underbrace{h, \varepsilon X f, u g, pp}_{=(27)} - \underbrace{h, \varepsilon X f, \varepsilon X g, pp}_{=-(28)} - \underbrace{h, \varepsilon X f, pp g, u}_{=-(29)} + \underbrace{h, \varepsilon X f, pp g, \varepsilon X}_{=(30)} - \underbrace{h, \varepsilon X f, p g, up}_{=-(31)} + \quad (56)$$

$$+ \underbrace{h, \varepsilon X f, p g, \varepsilon X p}_{=(32)} + \underbrace{g, uu h, p f, p}_{=(7)} - \underbrace{g, \varepsilon X u h, p f, p}_{=-(8)} + \underbrace{g, u h, pu f, p}_{=(33)} - \underbrace{g, \varepsilon X h, pu f, p}_{=-(34)} - \underbrace{g, pu h, u f, p}_{=-(23)} + \quad (57)$$

$$+ \underbrace{g, pu h, \varepsilon X f, p}_{=(31)} - \underbrace{g, p h, uu f, p}_{=-(35)} + \underbrace{g, p h, \varepsilon X u f, p}_{=(36)} - \underbrace{g, u \varepsilon X h, p f, p}_{=-(15)} + \underbrace{g, \varepsilon X \varepsilon X h, p f, p}_{=(16)} - \underbrace{g, u h, p \varepsilon X f, p}_{=-(37)} + \quad (58)$$

$$+ \underbrace{g, \varepsilon X h, p \varepsilon X f, p}_{=(38)} + \underbrace{g, p \varepsilon X h, u f, p}_{=(24)} - \underbrace{g, p \varepsilon X h, \varepsilon X f, p}_{=-(32)} + \underbrace{g, p h, u \varepsilon X f, p}_{=(39)} - \underbrace{g, p h, \varepsilon X \varepsilon X f, p}_{=-(40)} - \underbrace{f, u g, up h, p}_{=-(3)} + \quad (59)$$

$$+ \underbrace{f, u g, \varepsilon X p h, p}_{=(11)} - \underbrace{f, u g, u h, pp}_{=-(41)} + \underbrace{f, u g, \varepsilon X h, pp}_{=(42)} + \underbrace{f, u h, u g, pp}_{=(19)} - \underbrace{f, u g, pp h, \varepsilon X}_{=-(27)} + \underbrace{f, u g, p h, up}_{=(43)} - \underbrace{f, u g, p h, \varepsilon X p}_{=-(44)} + \quad (60)$$

$$+ \underbrace{g, up h, p f, \varepsilon X}_{=(4)} - \underbrace{g, \varepsilon X p h, p f, \varepsilon X}_{=(12)} + \underbrace{g, u h, pp f, \varepsilon X}_{=(45)} - \underbrace{g, \varepsilon X h, pp f, \varepsilon X}_{=-(46)} - \underbrace{h, u g, pp f, \varepsilon X}_{=-(20)} + \underbrace{g, pp h, \varepsilon X f, \varepsilon X}_{=(28)} + \quad (61)$$

$$- \underbrace{g, p h, up f, \varepsilon X}_{=-(47)} + \underbrace{g, p h, \varepsilon X p f, \varepsilon X}_{=(48)} + \underbrace{h, uu f, p g, p}_{=(35)} - \underbrace{h, \varepsilon X u f, p g, p}_{=-(36)} + \underbrace{h, u f, pu g, p}_{=(17)} - \underbrace{h, \varepsilon X f, pu g, p}_{=-(25)} + \quad (62)$$

$$\underbrace{-h_{puf,ug,p} + h_{puf,\varepsilon Xg,p} - h_{p f,uu g,p} + h_{p f,\varepsilon X u g,p} - h_{u\varepsilon X f,p g,p} + h_{\varepsilon X \varepsilon X f,p g,p}}_{=-(43)} + \underbrace{h_{puf,\varepsilon Xg,p}}_{=(47)} - \underbrace{h_{p f,uu g,p}}_{=-(1)} + \underbrace{h_{p f,\varepsilon X u g,p}}_{=(2)} - \underbrace{h_{u\varepsilon X f,p g,p}}_{=-(39)} + \underbrace{h_{\varepsilon X \varepsilon X f,p g,p}}_{=(40)} \quad (63)$$

$$\underbrace{-f_{p\varepsilon Xg,p}h_{u,p} + f_{p\varepsilon Xg,p}h_{\varepsilon X} + h_{p\varepsilon X f,u g,p} - h_{p\varepsilon X f,\varepsilon X g,p} + h_{p f,u\varepsilon Xg,p} - h_{p f,\varepsilon X \varepsilon X g,p}}_{=-(18)} + \underbrace{f_{p\varepsilon Xg,p}h_{\varepsilon X}}_{=(26)} + \underbrace{h_{p\varepsilon X f,u g,p}}_{=(44)} - \underbrace{h_{p\varepsilon X f,\varepsilon X g,p}}_{=-(48)} + \underbrace{h_{p f,u\varepsilon Xg,p}}_{=(9)} - \underbrace{h_{p f,\varepsilon X \varepsilon X g,p}}_{=-(10)} \quad (64)$$

$$\underbrace{-g_{u,p}h_{upf,p} + g_{u,p}h_{\varepsilon X p f,p} - g_{u,p}h_{u f,pp} + g_{u,p}h_{\varepsilon X f,pp} + g_{u,p}h_{ppf,u} - g_{u,p}h_{ppf,\varepsilon X} + g_{u,p}h_{p f,up}}_{=-(33)} + \underbrace{g_{u,p}h_{\varepsilon X p f,p}}_{=(37)} - \underbrace{g_{u,p}h_{u f,pp}}_{=-(21)} + \underbrace{g_{u,p}h_{\varepsilon X f,pp}}_{=(29)} + \underbrace{g_{u,p}h_{ppf,u}}_{=(41)} - \underbrace{g_{u,p}h_{ppf,\varepsilon X}}_{=-(45)} + \underbrace{g_{u,p}h_{p f,up}}_{=(5)} \quad (65)$$

$$\underbrace{-g_{u,p}h_{p f,\varepsilon X p} + h_{upf,p}g_{\varepsilon X} - h_{\varepsilon X p f,p}g_{\varepsilon X} + h_{u f,pp}g_{\varepsilon X} - h_{\varepsilon X f,pp}g_{\varepsilon X} - h_{ppf,u}g_{\varepsilon X}}_{=(13)} + \underbrace{h_{upf,p}g_{\varepsilon X}}_{=(34)} - \underbrace{h_{\varepsilon X p f,p}g_{\varepsilon X}}_{=-(38)} + \underbrace{h_{u f,pp}g_{\varepsilon X}}_{=(22)} - \underbrace{h_{\varepsilon X f,pp}g_{\varepsilon X}}_{=-(30)} - \underbrace{h_{ppf,u}g_{\varepsilon X}}_{=-(42)} \quad (66)$$

$$+ \underbrace{h_{ppf,\varepsilon Xg,\varepsilon X} - h_{p f,up}g_{\varepsilon X} + h_{p f,\varepsilon X p}g_{\varepsilon X}}_{=(46)} - \underbrace{h_{p f,up}g_{\varepsilon X}}_{=-(6)} + \underbrace{h_{p f,\varepsilon X p}g_{\varepsilon X}}_{=(14)} \quad (67)$$

$$=0 \quad (68)$$

Thermal - Thermal

We have

$$\{\{f, g\}_{\text{thermal}}, h\}_{\text{thermal}} = \{s_{\varepsilon} s_T^{-1} (g_{,T} f_{,pX} - f_{,T} g_{,pX}), h\}_{\text{thermal}} \quad (69)$$

$$= s_{\varepsilon} s_T^{-1} (h_{,T} (s_{\varepsilon} s_T^{-1} (g_{,T} f_{,pX} - f_{,T} g_{,pX})))_{,pX} - (s_{\varepsilon} s_T^{-1} (g_{,T} f_{,pX} - f_{,T} g_{,pX}))_{,T} h_{,pX} \quad (70)$$

$$= (s_{\varepsilon} s_T^{-1})^2 (h_{,T} g_{,T pX} f_{,pX} + h_{,T} g_{,T} f_{,pX pX} - h_{,T} f_{,T pX} g_{,pX} - h_{,T} f_{,T} g_{,pX pX}) + \quad (71)$$

$$- (s_{\varepsilon} s_T s_{,T} - s_{\varepsilon} s_{,TT}) (s_{,T})^{-2} (g_{,T} f_{,pX} h_{,pX} - f_{,T} g_{,pX} h_{,pX}) + \quad (72)$$

$$- (s_{\varepsilon} s_T^{-1})^2 (h_{,pX} g_{,TT} f_{,pX} + h_{,pX} g_{,T} f_{,pXT} - h_{,pX} f_{,TT} g_{,pX} - h_{,pX} f_{,T} g_{,pXT}) \quad (73)$$

Then, this part of the the Jacobi identity is

$$\{\{f, g\}_{\text{thermal}}, h\}_{\text{thermal}} + \{\{g, h\}_{\text{thermal}}, f\}_{\text{thermal}} + \{\{h, f\}_{\text{thermal}}, g\}_{\text{thermal}} \quad (74)$$

$$= (s_{\varepsilon} s_T^{-1})^2 (\underbrace{h_{,T} g_{,T pX} f_{,pX}}_{=(1)} + \underbrace{h_{,T} g_{,T} f_{,pX pX}}_{=(2)} - \underbrace{h_{,T} f_{,T pX} g_{,pX}}_{=-(3)} - \underbrace{h_{,T} f_{,T} g_{,pX pX}}_{=-(4)}) + \quad (75)$$

$$+ (s_{\varepsilon} s_T s_{,T} - s_{\varepsilon} s_{,TT}) (s_{,T})^{-2} (\underbrace{-g_{,T} f_{,pX} h_{,pX}}_{=-(5)} + \underbrace{f_{,T} g_{,pX} h_{,pX}}_{=(6)}) + \quad (76)$$

$$+ (s_{\varepsilon} s_T^{-1})^2 (\underbrace{-h_{,pX} g_{,TT} f_{,pX}}_{=-(8)} - \underbrace{h_{,pX} g_{,T} f_{,pXT}}_{=-(9)} + \underbrace{h_{,pX} f_{,TT} g_{,pX}}_{=(10)} + \underbrace{h_{,pX} f_{,T} g_{,pXT}}_{=(11)}) + \quad (77)$$

$$+ (s_{\varepsilon} s_T^{-1})^2 (\underbrace{f_{,T} h_{,T pX} g_{,pX}}_{=(12)} + \underbrace{f_{,T} h_{,T} g_{,pX pX}}_{=(4)} - \underbrace{f_{,T} g_{,T pX} h_{,pX}}_{=-(11)} - \underbrace{f_{,T} g_{,T} h_{,pX pX}}_{=-(13)}) + \quad (78)$$

$$+ (s_{\varepsilon} s_T s_{,T} - s_{\varepsilon} s_{,TT}) (s_{,T})^{-2} (\underbrace{-h_{,T} g_{,pX} f_{,pX}}_{=-(7)} + \underbrace{g_{,T} h_{,pX} f_{,pX}}_{=(5)}) + \quad (79)$$

$$+ (s_{\varepsilon} s_T^{-1})^2 (\underbrace{-f_{,pX} h_{,TT} g_{,pX}}_{=-(14)} - \underbrace{f_{,pX} h_{,T} g_{,pXT}}_{=-(1)} + \underbrace{f_{,pX} g_{,TT} h_{,pX}}_{=(8)} + \underbrace{f_{,pX} g_{,T} h_{,pXT}}_{=(15)}) + \quad (80)$$

$$+ (s_{\varepsilon} s_T^{-1})^2 (\underbrace{g_{,T} f_{,T pX} h_{,pX}}_{=(9)} + \underbrace{g_{,T} f_{,T} h_{,pX pX}}_{=(13)} - \underbrace{g_{,T} h_{,T pX} f_{,pX}}_{=-(15)} - \underbrace{g_{,T} h_{,T} f_{,pX pX}}_{=-(2)}) + \quad (81)$$

$$+ (s_{\varepsilon} s_T s_{,T} - s_{\varepsilon} s_{,TT}) (s_{,T})^{-2} (\underbrace{-f_{,T} h_{,pX} g_{,pX}}_{=-(6)} + \underbrace{h_{,T} f_{,pX} g_{,pX}}_{=(7)}) + \quad (82)$$

$$+ (s_{\varepsilon} s_T^{-1})^2 (\underbrace{-g_{,pX} f_{,TT} h_{,pX}}_{=-(10)} - \underbrace{g_{,pX} f_{,T} h_{,pXT}}_{=-(12)} + \underbrace{g_{,pX} h_{,TT} f_{,pX}}_{=(14)} + \underbrace{g_{,pX} h_{,T} f_{,pXT}}_{=(3)}) \quad (83)$$

Thermal-Canonical and Canonical-Thermal

First, we calculate the Canonical-Thermal bracket

$$\{\{f, g\}_{\text{canonical}}, h\}_{\text{thermal}} \quad (84)$$

$$= \{(f_{,u} - f_{,\varepsilon X})g_{,p} - f_{,p}(g_{,u} - g_{,\varepsilon X}), h\}_{\text{thermal}} \quad (85)$$

$$= s_{,\varepsilon} s_{,T}^{-1} (h_{,T} ((f_{,u} - f_{,\varepsilon X}) g_{,p} - f_{,p} (g_{,u} - g_{,\varepsilon X}))_{,pX} - h_{,pX} ((f_{,u} - f_{,\varepsilon X}) g_{,p} - f_{,p} (g_{,u} - g_{,\varepsilon X}))_{,T}) \quad (86)$$

$$= s_{,\varepsilon} s_{,T}^{-1} (h_{,T} f_{,upX} g_{,p} - h_{,T} f_{,\varepsilon X pX} g_{,p} + h_{,T} f_{,u g,ppX} - h_{,T} f_{,\varepsilon X g,ppX}) + \quad (87)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-h_{,T} f_{ppX} g_{,u} + h_{,T} f_{ppX} g_{,\varepsilon X} + h_{,T} f_{,p g,\varepsilon X pX} - h_{,T} f_{,p g,upX}) + \quad (88)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-h_{,pX} g_{,pT} f_{,u} + h_{,pX} g_{,pT} f_{,\varepsilon X} - h_{,pX} f_{,uT} g_{,p} + h_{,pX} f_{,\varepsilon XT} g_{,p}) + \quad (89)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (h_{,pX} f_{,pT} g_{,u} - h_{,pX} f_{,pT} g_{,\varepsilon X} + h_{,pX} f_{,p g,ut} - h_{,pX} f_{,p g,\varepsilon XT}) \quad (90)$$

$$\{\{g, h\}_{\text{canonical}}, f\}_{\text{thermal}} \quad (91)$$

$$= s_{,\varepsilon} s_{,T}^{-1} (f_{,T} g_{,upX} h_{,p} - f_{,T} g_{,\varepsilon X pX} h_{,p} + f_{,T} g_{,u h,ppX} - f_{,T} g_{,\varepsilon X h,ppX}) + \quad (92)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-f_{,T} g_{ppX} h_{,u} + f_{,T} g_{ppX} h_{,\varepsilon X} + f_{,T} g_{,p h,\varepsilon X pX} - f_{,T} g_{,p h,upX}) + \quad (93)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-f_{,pX} h_{,pT} g_{,u} + f_{,pX} h_{,pT} g_{,\varepsilon X} - f_{,pX} g_{,uT} h_{,p} + f_{,pX} g_{,\varepsilon XT} h_{,p}) + \quad (94)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (f_{,pX} g_{,pT} h_{,u} - f_{,pX} g_{,pT} h_{,\varepsilon X} + f_{,pX} g_{,p h,ut} - f_{,pX} g_{,p h,\varepsilon XT}) \quad (95)$$

$$\{\{h, f\}_{\text{canonical}}, g\}_{\text{thermal}} \quad (96)$$

$$= s_{,\varepsilon} s_{,T}^{-1} (g_{,T} h_{,upX} f_{,p} - g_{,T} h_{,\varepsilon X pX} f_{,p} + g_{,T} h_{,u f,ppX} - g_{,T} h_{,\varepsilon X f,ppX}) + \quad (97)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-g_{,T} h_{ppX} f_{,u} + g_{,T} h_{ppX} f_{,\varepsilon X} + g_{,T} h_{,p f,\varepsilon X pX} - g_{,T} h_{,p f,upX}) + \quad (98)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-g_{,pX} f_{,pT} h_{,u} + g_{,pX} f_{,pT} h_{,\varepsilon X} - g_{,pX} h_{,uT} f_{,p} + g_{,pX} h_{,\varepsilon XT} f_{,p}) + \quad (99)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (g_{,pX} h_{,pT} f_{,u} - g_{,pX} h_{,pT} f_{,\varepsilon X} + g_{,pX} h_{,p f,ut} - g_{,pX} h_{,p f,\varepsilon XT}) \quad (100)$$

Then, for the Thermal-Canonical brackets, we have

$$\{\{f, g\}_{\text{thermal}}, h\}_{\text{canonical}} = \{s_{,\varepsilon} s_{,T}^{-1} (g_{,T} f_{,pX} - f_{,T} g_{,pX}), h\}_{\text{canonical}} \quad (101)$$

$$= ((s_{,\varepsilon} s_{,T}^{-1} (g_{,T} f_{,pX} - f_{,T} g_{,pX}))_{,u} - (s_{,\varepsilon} s_{,T}^{-1} (g_{,T} f_{,pX} - f_{,T} g_{,pX}))_{,\varepsilon X}) h_{,p} + \quad (102)$$

$$- (s_{,\varepsilon} s_{,T}^{-1} (g_{,T} f_{,pX} - f_{,T} g_{,pX}))_{,p} (h_{,u} - h_{,\varepsilon X}) \quad (103)$$

$$= s_{,\varepsilon} s_{,T}^{-1} (g_{,Tu} f_{,pX} h_{,p} + g_{,T} f_{,pXu} h_{,p} - f_{,Tu} g_{,pX} h_{,p} - f_{,T} g_{,pXu} h_{,p}) + \quad (104)$$

$$+ (s_{,\varepsilon \varepsilon X} s_{,T} - s_{,\varepsilon} s_{,T \varepsilon X}) s_{,T}^{-2} (-g_{,T} f_{,pX} h_{,p} + f_{,T} g_{,pX} h_{,p}) + \quad (105)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-g_{,T \varepsilon X} f_{,pX} h_{,p} - g_{,T} f_{,pX \varepsilon X} h_{,p} + f_{,T \varepsilon X} g_{,pX} h_{,p} + f_{,T} g_{,pX \varepsilon X} h_{,p}) + \quad (106)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-g_{,Tp} f_{,pX} h_{,u} - g_{,T} f_{,pXp} h_{,u} + f_{,Tp} g_{,pX} h_{,u} + f_{,T} g_{,pXp} h_{,u}) + \quad (107)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (g_{,Tp} f_{,pX} h_{,\varepsilon X} + g_{,T} f_{,pXp} h_{,\varepsilon X} - f_{,Tp} g_{,pX} h_{,\varepsilon X} - f_{,T} g_{,pXp} h_{,\varepsilon X}) \quad (108)$$

$$\{\{g, h\}_{\text{thermal}}, f\}_{\text{canonical}} = s_{,\varepsilon} s_{,T}^{-1} (h_{,Tu} g_{,pX} f_{,p} + h_{,T} g_{,pXu} f_{,p} - g_{,Tu} h_{,pX} f_{,p} - g_{,T} h_{,pXu} f_{,p}) + \quad (109)$$

$$+ (s_{,\varepsilon \varepsilon X} s_{,T} - s_{,\varepsilon} s_{,T \varepsilon X}) s_{,T}^{-2} (-h_{,T} g_{,pX} f_{,p} + g_{,T} h_{,pX} f_{,p}) + \quad (110)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-h_{,T \varepsilon X} g_{,pX} f_{,p} - h_{,T} g_{,pX \varepsilon X} f_{,p} + g_{,T \varepsilon X} h_{,pX} f_{,p} + g_{,T} h_{,pX \varepsilon X} f_{,p}) + \quad (111)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-h_{,Tp} g_{,pX} f_{,u} - h_{,T} g_{,pXp} f_{,u} + g_{,Tp} h_{,pX} f_{,u} + g_{,T} h_{,pXp} f_{,u}) + \quad (112)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (h_{,Tp} g_{,pX} f_{,\varepsilon X} + h_{,T} g_{,pXp} f_{,\varepsilon X} - g_{,Tp} h_{,pX} f_{,\varepsilon X} - g_{,T} h_{,pXp} f_{,\varepsilon X}) \quad (113)$$

$$\{\{h, f\}_{\text{thermal}}, g\}_{\text{canonical}} = s_{,\varepsilon} s_{,T}^{-1} (f_{,Tu} h_{,pX} g_{,p} + f_{,T} h_{,pXu} g_{,p} - h_{,Tu} f_{,pX} g_{,p} - h_{,T} f_{,pXu} g_{,p}) + \quad (114)$$

$$+ (s_{,\varepsilon \varepsilon X} s_{,T} - s_{,\varepsilon} s_{,T \varepsilon X}) s_{,T}^{-2} (-f_{,T} h_{,pX} g_{,p} + h_{,T} f_{,pX} g_{,p}) + \quad (115)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-f_{,T \varepsilon X} h_{,pX} g_{,p} - f_{,T} h_{,pX \varepsilon X} g_{,p} + h_{,T \varepsilon X} f_{,pX} g_{,p} + h_{,T} f_{,pX \varepsilon X} g_{,p}) + \quad (116)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (-f_{,Tp} h_{,pX} g_{,u} - f_{,T} h_{,pXp} g_{,u} + h_{,Tp} f_{,pX} g_{,u} + h_{,T} f_{,pXp} g_{,u}) + \quad (117)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} (f_{,Tp} h_{,pX} g_{,\varepsilon X} + f_{,T} h_{,pXp} g_{,\varepsilon X} - h_{,Tp} f_{,pX} g_{,\varepsilon X} - h_{,T} f_{,pXp} g_{,\varepsilon X}) \quad (118)$$

Therefore, we have

$$\{\{f, g\}_{\text{canonical}}, h\}_{\text{thermal}} + \{\{f, g\}_{\text{thermal}}, h\}_{\text{canonical}} + \{\{g, h\}_{\text{canonical}}, f\}_{\text{thermal}} + \quad (119)$$

$$+ \{\{g, h\}_{\text{thermal}}, f\}_{\text{canonical}} + \{\{h, f\}_{\text{canonical}}, g\}_{\text{thermal}} + \{\{h, f\}_{\text{thermal}}, g\}_{\text{canonical}} \quad (120)$$

$$= s_{,\varepsilon} s_{,T}^{-1} \underbrace{(h_{,T} f_{,upX} g_{,p} - h_{,T} f_{,\varepsilon X pX} g_{,p})}_{=(1)} + \underbrace{(h_{,T} f_{,u g,ppX} - h_{,T} f_{,\varepsilon X g,ppX})}_{=(3)} + \underbrace{(h_{,T} f_{,p g,\varepsilon X pX} - h_{,T} f_{,p g,upX})}_{=-(4)} + \quad (121)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-h_{,T} f_{ppX} g_{,u} + h_{,T} f_{ppX} g_{,\varepsilon X})}_{=-(5)} + \underbrace{(h_{,T} f_{,p g,\varepsilon X pX} - h_{,T} f_{,p g,upX})}_{=(6)} + \underbrace{(h_{,T} f_{,p g,\varepsilon X pX} - h_{,T} f_{,p g,upX})}_{=(7)} + \underbrace{(h_{,T} f_{,p g,\varepsilon X pX} - h_{,T} f_{,p g,upX})}_{=-(8)} \quad (122)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-h_{,pX} g_{,pT} f_{,u})}_{=-(9)} + \underbrace{h_{,pX} g_{,pT} f_{,\varepsilon X}}_{=(10)} - \underbrace{h_{,pX} f_{,uT} g_{,p}}_{=-(11)} + \underbrace{h_{,pX} f_{,\varepsilon XT} g_{,p}}_{=(12)} + \quad (123)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(h_{,pX} f_{,pT} g_{,u})}_{=(13)} - \underbrace{h_{,pX} f_{,pT} g_{,\varepsilon X}}_{=-(14)} + \underbrace{h_{,pX} f_{,p} g_{,uT}}_{=(15)} - \underbrace{h_{,pX} f_{,p} g_{,\varepsilon XT}}_{=-(16)} + \quad (124)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(f_{,T} g_{,u} p_X h_{,p})}_{=(17)} - \underbrace{f_{,T} g_{,\varepsilon X} p_X h_{,p}}_{=-(18)} + \underbrace{f_{,T} g_{,u} h_{,ppX}}_{=(19)} - \underbrace{f_{,T} g_{,\varepsilon X} h_{,ppX}}_{=-(20)} + \quad (125)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-f_{,T} g_{,ppX} h_{,u})}_{=-(21)} + \underbrace{f_{,T} g_{,ppX} h_{,\varepsilon X}}_{=(22)} + \underbrace{f_{,T} g_{,p} h_{,\varepsilon X} p_X}_{=(23)} - \underbrace{f_{,T} g_{,p} h_{,u} p_X}_{=-(24)} + \quad (126)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-f_{,pX} h_{,pT} g_{,u})}_{=-(25)} + \underbrace{f_{,pX} h_{,pT} g_{,\varepsilon X}}_{=(26)} - \underbrace{f_{,pX} g_{,uT} h_{,p}}_{=-(27)} + \underbrace{f_{,pX} g_{,\varepsilon XT} h_{,p}}_{=(28)} + \quad (127)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(f_{,pX} g_{,pT} h_{,u})}_{=(29)} - \underbrace{f_{,pX} g_{,pT} h_{,\varepsilon X}}_{=-(30)} + \underbrace{f_{,pX} g_{,p} h_{,uT}}_{=(31)} - \underbrace{f_{,pX} g_{,p} h_{,\varepsilon XT}}_{=-(32)} + \quad (128)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(g_{,T} h_{,u} p_X f_{,p})}_{=(33)} - \underbrace{g_{,T} h_{,\varepsilon X} p_X f_{,p}}_{=-(34)} + \underbrace{g_{,T} h_{,u} f_{,ppX}}_{=(35)} - \underbrace{g_{,T} h_{,\varepsilon X} f_{,ppX}}_{=-(36)} + \quad (129)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-g_{,T} h_{,ppX} f_{,u})}_{=-(37)} + \underbrace{g_{,T} h_{,ppX} f_{,\varepsilon X}}_{=(38)} + \underbrace{g_{,T} h_{,p} f_{,\varepsilon X} p_X}_{=(39)} - \underbrace{g_{,T} h_{,p} f_{,u} p_X}_{=-(40)} + \quad (130)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-g_{,pX} f_{,pT} h_{,u})}_{=-(41)} + \underbrace{g_{,pX} f_{,pT} h_{,\varepsilon X}}_{=(42)} - \underbrace{g_{,pX} h_{,uT} f_{,p}}_{=-(43)} + \underbrace{g_{,pX} h_{,\varepsilon XT} f_{,p}}_{=(44)} + \quad (131)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(g_{,pX} h_{,pT} f_{,u})}_{=(45)} - \underbrace{g_{,pX} h_{,pT} f_{,\varepsilon X}}_{=-(46)} + \underbrace{g_{,pX} h_{,p} f_{,uT}}_{=(47)} - \underbrace{g_{,pX} h_{,p} f_{,\varepsilon XT}}_{=-(48)} + \quad (132)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(g_{,Tu} f_{,pX} h_{,p})}_{=(27)} + \underbrace{g_{,T} f_{,pXu} h_{,p}}_{=(40)} - \underbrace{f_{,Tu} g_{,pX} h_{,p}}_{=-(47)} - \underbrace{f_{,T} g_{,pXu} h_{,p}}_{=-(17)} + \quad (133)$$

$$+ (s_{,\varepsilon\varepsilon X} s_{,T} - s_{,\varepsilon} s_{,T\varepsilon X}) s_{,T}^{-2} (-g_{,T} f_{,pX} h_{,p} + f_{,T} g_{,pX} h_{,p}) + \quad (134)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-g_{,T\varepsilon X} f_{,pX} h_{,p})}_{=-(28)} - \underbrace{g_{,T} f_{,pX\varepsilon X} h_{,p}}_{=-(39)} + \underbrace{f_{,T\varepsilon X} g_{,pX} h_{,p}}_{=(48)} + \underbrace{f_{,T} g_{,pX\varepsilon X} h_{,p}}_{=(18)} + \quad (135)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-g_{,Tp} f_{,pX} h_{,u})}_{=-(29)} - \underbrace{g_{,T} f_{,pXp} h_{,u}}_{=-(35)} + \underbrace{f_{,Tp} g_{,pX} h_{,u}}_{=(41)} + \underbrace{f_{,T} g_{,pXp} h_{,u}}_{=(21)} + \quad (136)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(g_{,Tp} f_{,pX} h_{,\varepsilon X})}_{=(30)} + \underbrace{g_{,T} f_{,pXp} h_{,\varepsilon X}}_{=(36)} - \underbrace{f_{,Tp} g_{,pX} h_{,\varepsilon X}}_{=-(42)} - \underbrace{f_{,T} g_{,pXp} h_{,\varepsilon X}}_{=-(22)} + \quad (137)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(h_{,Tu} g_{,pX} f_{,p})}_{=(43)} + \underbrace{h_{,T} g_{,pXu} f_{,p}}_{=(8)} - \underbrace{g_{,Tu} h_{,pX} f_{,p}}_{=-(15)} - \underbrace{g_{,T} h_{,pXu} f_{,p}}_{=-(33)} + \quad (138)$$

$$+ (s_{,\varepsilon\varepsilon X} s_{,T} - s_{,\varepsilon} s_{,T\varepsilon X}) s_{,T}^{-2} (-h_{,T} g_{,pX} f_{,p} + g_{,T} h_{,pX} f_{,p}) + \quad (139)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-h_{,T\varepsilon X} g_{,pX} f_{,p})}_{=-(44)} - \underbrace{h_{,T} g_{,pX\varepsilon X} f_{,p}}_{=-(7)} + \underbrace{g_{,T\varepsilon X} h_{,pX} f_{,p}}_{=(16)} + \underbrace{g_{,T} h_{,pX\varepsilon X} f_{,p}}_{=(34)} + \quad (140)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-h_{,Tp} g_{,pX} f_{,u})}_{=-(45)} - \underbrace{h_{,T} g_{,pXp} f_{,u}}_{=-(3)} + \underbrace{g_{,Tp} h_{,pX} f_{,u}}_{=(9)} + \underbrace{g_{,T} h_{,pXp} f_{,u}}_{=(37)} + \quad (141)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(h_{,Tp} g_{,pX} f_{,\varepsilon X})}_{=(46)} + \underbrace{h_{,T} g_{,pXp} f_{,\varepsilon X}}_{=(4)} - \underbrace{g_{,Tp} h_{,pX} f_{,\varepsilon X}}_{=-(10)} - \underbrace{g_{,T} h_{,pXp} f_{,\varepsilon X}}_{=-(38)} + \quad (142)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(f_{,Tu} h_{,pX} g_{,p})}_{=(11)} + \underbrace{f_{,T} h_{,pXu} g_{,p}}_{=(24)} - \underbrace{h_{,Tu} f_{,pX} g_{,p}}_{=-(31)} - \underbrace{h_{,T} f_{,pXu} g_{,p}}_{=-(1)} + \quad (143)$$

$$+ (s_{,\varepsilon\varepsilon X} s_{,T} - s_{,\varepsilon} s_{,T\varepsilon X}) s_{,T}^{-2} (-f_{,T} h_{,pX} g_{,p} + h_{,T} f_{,pX} g_{,p}) + \quad (144)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-f_{,T\varepsilon X} h_{,pX} g_{,p})}_{=-(12)} - \underbrace{f_{,T} h_{,pX\varepsilon X} g_{,p}}_{=-(23)} + \underbrace{h_{,T\varepsilon X} f_{,pX} g_{,p}}_{=(32)} + \underbrace{h_{,T} f_{,pX\varepsilon X} g_{,p}}_{=(2)} + \quad (145)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(-f_{,Tp} h_{,pX} g_{,u})}_{=-(13)} - \underbrace{f_{,T} h_{,pXp} g_{,u}}_{=-(19)} + \underbrace{h_{,Tp} f_{,pX} g_{,u}}_{=(25)} + \underbrace{h_{,T} f_{,pXp} g_{,u}}_{=(5)} + \quad (146)$$

$$+ s_{,\varepsilon} s_{,T}^{-1} \underbrace{(f_{,Tp} h_{,pX} g_{,\varepsilon X})}_{=(14)} + \underbrace{f_{,T} h_{,pXp} g_{,\varepsilon X}}_{=(20)} - \underbrace{h_{,Tp} f_{,pX} g_{,\varepsilon X}}_{=-(26)} - \underbrace{h_{,T} f_{,pXp} g_{,\varepsilon X}}_{=-(6)} + \quad (147)$$

$$= (s_{,\varepsilon\varepsilon X} s_{,T} - s_{,\varepsilon} s_{,T\varepsilon X}) s_{,T}^{-2} (-g_{,T} f_{,pX} h_{,p} + f_{,T} g_{,pX} h_{,p} - h_{,T} g_{,pX} f_{,p} + \quad (148)$$

$$+ g_{,T} h_{,pX} f_{,p} - f_{,T} h_{,pX} g_{,p} + h_{,T} f_{,pX} g_{,p}) \quad (149)$$

We thus need the following conditions to satisfy the the Jacobi identity

$$\frac{\partial^3 s}{\partial \varepsilon^2 \partial X} = \frac{\partial}{\partial X} \left(\frac{\partial^2 s}{\partial \varepsilon^2} \right) = 0 \quad (150)$$

$$\frac{\partial^3 s}{\partial T \partial \varepsilon \partial X} = \frac{\partial}{\partial X} \left(\frac{\partial^2 s}{\partial T \partial \varepsilon} \right) = 0 \quad (151)$$

These conditions are physically reasonable. Indeed, considering the expression of the entropy for linear thermo-elasticity

$$s(\varepsilon, T) = s_0 + \frac{k}{\rho} \varepsilon + c \frac{T}{T_0} \quad (152)$$

the previous conditions are automatically fulfilled.

_____ *qed.* ■

1.5. Symmetry of the dissipative bracket

The symmetry of the dissipative bracket is verified in the following calculus

$$(\mathcal{F}, \mathcal{S}) - (\mathcal{S}, \mathcal{F}) \quad (153)$$

$$= \int_{\omega_X} \left[\underbrace{\frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) K \frac{T^2}{c^2} \frac{\partial}{\partial X} \left(\frac{\partial s}{\partial T} \right)}_{(a)} + \underbrace{\frac{\partial f}{\partial T} \frac{1}{c} \frac{T}{c} \left(\frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \right)^\top \mathbf{V}^{-1} \frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \frac{\partial s}{\partial T}}_{=(b)} + \right. \quad (154)$$

$$\left. - \frac{\partial f}{\partial T} \frac{T}{c} \left(\frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \right)^\top \mathbf{V}^{-1} \frac{\partial s}{\partial \alpha} - \frac{T}{c} \left(\frac{\partial f}{\partial \alpha} \right)^\top \mathbf{V}^{-1} \frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \frac{\partial s}{\partial T} + T \left(\frac{\partial f}{\partial \alpha} \right)^\top \mathbf{V}^{-1} \frac{\partial s}{\partial \alpha} + \right. \quad (155)$$

$$\left. - \frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) K \frac{T^2}{c^2} \frac{\partial}{\partial X} \left(\frac{\partial s}{\partial T} \right) - \frac{\partial f}{\partial T} \frac{1}{c} \frac{T}{c} \left(\frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \right)^\top \mathbf{V}^{-1} \frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \frac{\partial s}{\partial T} + \right. \quad (156)$$

$$\left. + \frac{\partial s}{\partial T} \frac{T}{c} \left(\frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \right)^\top \mathbf{V}^{-1} \frac{\partial f}{\partial \alpha} + \frac{T}{c} \left(\frac{\partial s}{\partial \alpha} \right)^\top \mathbf{V}^{-1} \frac{\partial \hat{e}_{\text{int}}}{\partial \alpha} \frac{\partial f}{\partial T} - T \left(\frac{\partial s}{\partial \alpha} \right)^\top \mathbf{V}^{-1} \frac{\partial f}{\partial \alpha} \right] d\omega_X \quad (157)$$

$$= 0 \quad (158)$$

due to the symmetry of $\mathbf{V}^{-1}$.

2. Tri-dimensional large strain thermo-visco-elasticity

The Double Bracket Structure for large strain thermo-visco-elastodynamics is shown to be equal to

$$\frac{d\mathcal{F}}{dt} = \int_{\omega_X} \left(\frac{\partial f}{\partial \underline{\chi}} \cdot \frac{\partial \underline{\chi}}{\partial t} + \frac{\partial f}{\partial \underline{p}} \cdot \frac{\partial \underline{p}}{\partial t} - \frac{\partial f}{\partial \underline{F}} : \dot{\underline{F}} + \frac{\partial f}{\partial T} \left(\left(\frac{\partial T}{\partial t} \right)_{\text{rev}} + \left(\frac{\partial T}{\partial t} \right)_{\text{irr}} \right) + \frac{\partial f}{\partial \underline{C}_{\text{i}}^{-1}} : \frac{\partial \underline{C}_{\text{i}}^{-1}}{\partial t} \right) \quad (159)$$

$$= \{\mathcal{F}, \hat{\mathcal{E}}_{\text{tot}}\} + (\mathcal{F}, \mathcal{S}) + \quad (160)$$

$$+ \int_{\partial\omega_X} \frac{\partial f}{\partial \underline{p}} \cdot (\underline{\Pi} \cdot \underline{N}) d(\partial\omega_X) - \int_{\partial\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} K \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial \underline{X}} \left(\frac{\partial s}{\partial T} \right) \cdot \underline{N} d(\partial\omega_X) \quad (161)$$

where

$$\{\mathcal{F}, \hat{\mathcal{E}}_{\text{tot}}\} = \int_{\omega_X} \left(\frac{\partial f}{\partial \underline{\chi}} \cdot \frac{\partial \hat{\mathcal{E}}_{\text{tot}}^*}{\partial \underline{p}} - \frac{\partial f}{\partial \underline{p}} \cdot \frac{\partial \hat{\mathcal{E}}_{\text{tot}}^*}{\partial \underline{\chi}} + \frac{\partial f}{\partial \underline{F}} : \frac{\partial \hat{\mathcal{E}}_{\text{tot}}^*}{\partial \underline{\Pi}} \underline{S} - \frac{\partial f}{\partial \underline{\Pi}} : \frac{\partial \hat{\mathcal{E}}_{\text{tot}}^*}{\partial \underline{X}} \underline{S} + \right. \quad (162)$$

$$\left. - \frac{\partial \hat{\mathcal{E}}_{\text{tot}}^*}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \underline{\nabla} \frac{\partial f}{\partial \underline{p}} : \frac{\partial s^*}{\partial \underline{\Pi}} \underline{S} + \frac{\partial f}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \frac{\partial s^*}{\partial \underline{\Pi}} : \underline{\nabla} \frac{\partial \hat{\mathcal{E}}_{\text{tot}}^*}{\partial \underline{p}} \underline{S} \right) d\omega_X \quad (163)$$

$$\begin{aligned} (\mathcal{F}, \mathcal{S}) = & \int_{\omega_X} \left[\frac{1}{\rho} \frac{\partial}{\partial \underline{X}} \left(\frac{\partial f}{\partial T} \right) \cdot K \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial \underline{X}} \left(\frac{\partial s^*}{\partial T} \right) + \frac{\partial f}{\partial T} \frac{4}{c^2} \left(\frac{\partial \hat{\mathcal{E}}_{\text{int}}^*}{\partial \underline{C}_{\text{i}}^{-1}} \cdot \underline{C}_{\text{i}}^{-1} \right) : T \underline{N} : \left(\frac{\partial \hat{\mathcal{E}}_{\text{int}}^*}{\partial \underline{C}_{\text{i}}^{-1}} \cdot \underline{C}_{\text{i}}^{-1} \right) \frac{\partial s^*}{\partial T} + \right. \\ & - \frac{\partial f}{\partial T} \frac{4}{c} \left(\frac{\partial s^*}{\partial \underline{C}_{\text{i}}^{-1}} \cdot \underline{C}_{\text{i}}^{-1} \right) : T \underline{N} : \left(\frac{\partial \hat{\mathcal{E}}_{\text{int}}^*}{\partial \underline{C}_{\text{i}}^{-1}} \cdot \underline{C}_{\text{i}}^{-1} \right) - \frac{\partial s^*}{\partial T} \frac{4}{c} \left(\frac{\partial \hat{\mathcal{E}}_{\text{int}}^*}{\partial \underline{C}_{\text{i}}^{-1}} \cdot \underline{C}_{\text{i}}^{-1} \right) : T \underline{N} : \left(\frac{\partial f}{\partial \underline{C}_{\text{i}}^{-1}} \cdot \underline{C}_{\text{i}}^{-1} \right) + \\ & \left. + 4 \left(\frac{\partial s^*}{\partial \underline{C}_{\text{i}}^{-1}} \cdot \underline{C}_{\text{i}}^{-1} \right) : T \underline{N} : \left(\frac{\partial f}{\partial \underline{C}_{\text{i}}^{-1}} \cdot \underline{C}_{\text{i}}^{-1} \right) \right] d\omega_X \quad (164) \end{aligned}$$

2.1. Derivation of the viscoelastic law

Proposition 2.1

The evolution law of the inverse Cauchy Green tensor is given by [Reese and Govindjee 1998; Schiebl and Betsch 2021]

$$\frac{\partial \underline{C}_{\text{i}}^{-1}}{\partial t} = -4 \underline{F}^{-1} \cdot \left[\underline{V}^{-1} : \left(\underline{F}^{-\text{T}} \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{C}_{\text{i}}^{-1}} \cdot \underline{C}_{\text{i}}^{-1} \cdot \underline{F}^{\text{T}} \right) \right] \cdot \underline{F} \cdot \underline{C}_{\text{i}}^{-1} \quad (165)$$

Proof _____

The developments are based on the works of [Reese and Govindjee 1998]. We consider a general Helmholtz free energy density of the form

$$w = w(\underline{C}, \underline{Q}_1, \dots, \underline{Q}_k, T) \quad (166)$$

where $\underline{Q}_1, \dots, \underline{Q}_k$ are objective internal variables, and $\underline{C}$ is the Cauchy-Green tensor, so that the function is objective [Coleman and Gurtin 1967]. The mechanical dissipation $\mathcal{D}_{\text{mech}}$, which arises from the second law of thermodynamics, must remain positive

$$\mathcal{D}_{\text{mech}} = \underline{S} : \dot{\underline{E}} - \rho(\dot{T}s + \dot{w}) \quad (167)$$

$$= \left(\underline{S} - 2\rho \frac{\partial w}{\partial \underline{C}} \right) : \dot{\underline{E}} + \left(-\rho s - \rho \frac{\partial w}{\partial T} \right) : \dot{T} - \rho \sum_{i=1}^k \frac{\partial w}{\partial \underline{Q}_i} : \dot{\underline{Q}}_i \quad (168)$$

This equation gives the definition of the entropy and the second Piola Kirchhoff stress tensor in terms of the derivatives of Helmholtz free energy

$$s = -\frac{\partial w}{\partial T}, \quad \underline{\underline{S}} = 2\rho \frac{\partial w}{\partial \underline{\underline{C}}} \quad (169)$$

We take a density of Helmholtz free energy to make the notations lighter. Finite viscoelastic models are based on the decomposition of the Helmholtz free energy

$$w = w_{\text{EQ}}(\underline{\underline{C}}) + w_{\text{NEQ}}(\underline{\underline{C}}) \quad (170)$$

where $\underline{\underline{C}} = \underline{\underline{F}}^T \cdot \underline{\underline{F}}$ is the elastic Cauchy-Green tensor. This assumption has been verified experimentally [Bergstrom 1998]. Thanks to the Sidoroff decomposition, the Cauchy-Green tensor can therefore be expressed as

$$\underline{\underline{C}} = \underline{\underline{F}}^T \cdot \underline{\underline{F}} = (\underline{\underline{F}} \cdot \underline{\underline{F}}_i)^T \cdot (\underline{\underline{F}} \cdot \underline{\underline{F}}_i) = \underline{\underline{F}}_i^T \cdot \underline{\underline{F}}_e^T \cdot \underline{\underline{F}}_e \cdot \underline{\underline{F}}_i \quad (171)$$

$$\Rightarrow \underline{\underline{F}}_i^{-T} \cdot \underline{\underline{C}} \cdot \underline{\underline{F}}_i^{-1} = \underline{\underline{C}}_e \quad (172)$$

Moreover, due to the specific form of the Helmholtz free energy above, the internal objective variables $\underline{\underline{Q}}_i$ only represents the inelastic deformations, such that $\underline{\underline{Q}}_j = \underline{\underline{F}}_i$, and

$$\frac{\partial w}{\partial \underline{\underline{Q}}_j} : \dot{\underline{\underline{Q}}}_j = \frac{\partial w}{\partial \underline{\underline{F}}_i} : \frac{\partial \underline{\underline{F}}_i}{\partial t} = \frac{\partial w}{\partial \underline{\underline{C}}_e} : \frac{\partial \underline{\underline{C}}_e}{\partial \underline{\underline{F}}_i} : \frac{\partial \underline{\underline{F}}_i}{\partial t} \quad (173)$$

Thus, the dissipation inequality (168) becomes

$$\left(\underline{\underline{S}} - 2 \frac{\partial w_{\text{EQ}}}{\partial \underline{\underline{C}}} - 2 \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}} \right) : \frac{1}{2} \dot{\underline{\underline{C}}} - \frac{\partial w}{\partial \underline{\underline{C}}_e} : \frac{\partial \underline{\underline{C}}_e}{\partial \underline{\underline{F}}_i} : \frac{\partial \underline{\underline{F}}_i}{\partial t} \geq 0 \quad (174)$$

The derivative of the inverse of a matrix is given by the following calculus

$$\frac{\partial \underline{\underline{A}}^{-1}}{\partial \underline{\underline{A}}} = -\underline{\underline{A}} \otimes \underline{\underline{A}}, \quad \frac{\partial \underline{\underline{A}}_{ip}^{-1}}{\partial \underline{\underline{A}}_{mn}} = \underline{\underline{A}}_{im}^{-1} \underline{\underline{A}}_{np}^{-1} \quad (175)$$

Along with the relationship

$$\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}} = \underline{\underline{F}}_i^{-1} \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_e} \cdot \underline{\underline{F}}_i^{-T} \quad (176)$$

which is proved as

$$\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}} = \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_e} : \frac{\partial \underline{\underline{C}}_e}{\partial \underline{\underline{C}}} \quad (177)$$

In index, the expression yields

$$\left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}} \right)_{AB} = \left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_e} \right)_{PQ} \left(\frac{\partial \underline{\underline{C}}_e}{\partial \underline{\underline{C}}} \right)_{PABQ} = \left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_e} \right)_{PQ} \frac{\partial (\underline{\underline{F}}_i^{-T} \cdot \underline{\underline{C}} \cdot \underline{\underline{F}}_i^{-1})_{PQ}}{\partial (C)_{AB}} \quad (178)$$

$$= \left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_e} \right)_{PQ} \frac{\partial (\underline{\underline{F}}_i^{-1})_{JP} (C)_{JI} (\underline{\underline{F}}_i^{-1})_{IQ}}{\partial (C)_{AB}} = \left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_e} \right)_{PQ} (\underline{\underline{F}}_i^{-1})_{JP} \frac{\partial (C)_{JI}}{\partial (C)_{AB}} (\underline{\underline{F}}_i^{-1})_{IQ} \quad (179)$$

$$= \left(\frac{\partial w_{NEQ}}{\partial C_e} \right)_{PQ} (F_i^{-1})_{JP} \delta_{JA} \delta_{IB} (F_i^{-1})_{IQ} = \left(\frac{\partial w_{NEQ}}{\partial C_e} \right)_{PQ} (F_i^{-1})_{AP} (F_i^{-1})_{BQ} \quad (180)$$

$$= (F_i^{-1})_{AP} \left(\frac{\partial w_{NEQ}}{\partial C_e} \right)_{PQ} (F_i^{-T})_{QB} \quad (181)$$

We consider the inelastic part of the gradient of transformation to be purely Lagrangian, making its elastic part mixed (see **sadik-2024**). Therefore, one obtains

$$\frac{\partial w_{NEQ}}{\partial \underline{\underline{C}}} = \underline{\underline{F}}^{-1} \cdot \frac{\partial w_{NEQ}}{\partial \underline{\underline{C}}_e} \cdot \underline{\underline{F}}^{-T} \quad (182)$$

Then, the dissipation inequality becomes

$$\left(\underline{\underline{S}} - 2 \frac{\partial w_{EQ}}{\partial \underline{\underline{C}}} - 2 \underline{\underline{F}}^{-1} \cdot \frac{\partial w_{NEQ}}{\partial \underline{\underline{C}}_e} \cdot \underline{\underline{F}}^{-T} \right) : \frac{1}{2} \dot{\underline{\underline{C}}} - \frac{\partial w}{\partial \underline{\underline{C}}} : \frac{\partial \underline{\underline{C}}_e}{\partial \underline{\underline{F}}_i} : \frac{\partial \underline{\underline{F}}_i}{\partial t} \geq 0 \quad (183)$$

Following the work of Coleman and Gurtin [Coleman and Gurtin 1967], the second Piola Kirchhoff stress tensor may be split into an equilibrium and a non equilibrium part

$$\underline{\underline{S}} = \underline{\underline{S}}_{EQ} + \underline{\underline{S}}_{NEQ} = 2 \frac{\partial w_{EQ}}{\partial \underline{\underline{C}}} + 2 \underline{\underline{F}}^{-1} \cdot \frac{\partial w_{NEQ}}{\partial \underline{\underline{C}}_e} \cdot \underline{\underline{F}}^{-T} = 2 \frac{\partial w}{\partial \underline{\underline{C}}} \quad (184)$$

We thus consider that the Helmholtz free energy and the stress tensors are chosen such that the above relationship holds true. Then, the dissipation inequality is reduced to

$$- \frac{\partial w}{\partial \underline{\underline{C}}_e} : \frac{\partial \underline{\underline{C}}_e}{\partial \underline{\underline{F}}_i} : \frac{\partial \underline{\underline{F}}_i}{\partial t} \geq 0 \quad (185)$$

Then, we show that

$$- \frac{\partial w_{NEQ}}{\partial \underline{\underline{C}}_e} : \frac{\partial \underline{\underline{C}}_e}{\partial \underline{\underline{F}}_i} : \frac{\partial \underline{\underline{F}}_i}{\partial t} = 2 \frac{\partial w_{NEQ}}{\partial \underline{\underline{C}}_e} : \left(\underline{\underline{C}}_e \cdot \frac{\partial \underline{\underline{F}}_i}{\partial t} \cdot \underline{\underline{F}}_i^{-1} \right) \quad (186)$$

We first express the derivative of the elastic part of the Cauchy-Green tensor with respect to the inelastic gradient of transformation $\partial \underline{\underline{C}}_e / \partial \underline{\underline{F}}_i$, in components

$$\left(\frac{\partial \underline{\underline{C}}_e}{\partial \underline{\underline{F}}_i} \right)_{APQB} = \frac{\partial (C_e)_{AB}}{\partial (F_i)_{PQ}} = \frac{\partial (\underline{\underline{F}}_i^{-T} \cdot \underline{\underline{C}}_e \cdot \underline{\underline{F}}_i^{-1})_{AB}}{\partial (F_i)_{PQ}} = \frac{\partial (F_i^{-1})_{JA} C_{JI} (F_i^{-1})_{IB}}{\partial (F_i)_{PQ}} \quad (187)$$

$$= \frac{\partial (F_i^{-1})_{JA}}{\partial (F_i)_{PQ}} C_{JI} (F_i^{-1})_{IB} + (F_i^{-1})_{JA} C_{JI} \frac{\partial (F_i^{-1})_{IB}}{\partial (F_i)_{PQ}} \quad (188)$$

$$= -(F_i^{-1})_{JP} (F_i^{-1})_{QA} (C)_{JI} (F_i^{-1})_{IB} - (F_i^{-1})_{JA} (C)_{JI} (F_i^{-1})_{IP} (F_i^{-1})_{QB} \quad (189)$$

$$= -(F_i^{-1})_{QA} (\underline{\underline{F}}_i^{-T} \cdot \underline{\underline{C}}_e \cdot \underline{\underline{F}}_i^{-1})_{PB} - (F_i^{-1})_{QB} (\underline{\underline{F}}_i^{-T} \cdot \underline{\underline{C}}_e \cdot \underline{\underline{F}}_i^{-1})_{AP} \quad (190)$$

$$= -(F_i^{-1})_{QA} (C_e)_{PB} - (F_i^{-1})_{QB} (C_e)_{AP} \quad (191)$$

Then, we take the double contraction of this fourth order tensor with the time derivative of the inelastic gradient of transformations to obtain

$$\left(\frac{\partial \underline{\underline{C}}_e}{\partial \underline{\underline{F}}_i} \right)_{APQB} \left(\frac{\partial \underline{\underline{F}}_i}{\partial t} \right)_{PQ} = -(F_i^{-1})_{QA} (C_e)_{PB} \left(\frac{\partial \underline{\underline{F}}_i}{\partial t} \right)_{PQ} - (F_i^{-1})_{QB} (C_e)_{AP} \left(\frac{\partial \underline{\underline{F}}_i}{\partial t} \right)_{PQ} \quad (192)$$

$$= -(C_e^\top)_{BP} \left(\frac{\partial F_i}{\partial t} \right)_{PQ} (F_i^{-1})_{QA} - (C_e)_{AP} \left(\frac{\partial F_i}{\partial t} \right)_{PQ} (F_i^{-1})_{QB} \quad (193)$$

thus giving

$$\frac{\partial C}{\partial F} : \frac{\partial F}{\partial t} = -C_e^\top \cdot \frac{\partial F}{\partial t} \cdot F^{-1} - C_e \cdot \frac{\partial F}{\partial t} \cdot F^{-1} \quad (194)$$

$$= -2C_e \cdot \frac{\partial F}{\partial t} \cdot F^{-1} \quad (195)$$

where we used the symmetry of the elastic Cauchy Green tensor

$$\underline{C}_e^\top = \underline{C}_e \quad (196)$$

thus proving (186). We define the inelastic part of the spatial velocity gradient

$$\underline{l} = \frac{\partial F}{\partial t} \cdot F^{-1} \quad (197)$$

which is introduced in the dissipation inequality

$$2 \frac{\partial w_{\text{NEQ}}}{\partial C_e} : (C_e \cdot \frac{\partial F}{\partial t} \cdot F^{-1}) \geq 0 \Leftrightarrow 2 \frac{\partial w_{\text{NEQ}}}{\partial C_e} \cdot F^\top : (F^{-\top} \cdot C_e \cdot \underline{l} \cdot F^{-1}) \geq 0 \quad (198)$$

$$\Leftrightarrow 2 \frac{\partial w_{\text{NEQ}}}{\partial C_e} \cdot F^\top : (F \cdot \underline{l} \cdot F^{-1}) \geq 0 \quad (199)$$

Define the split of the Kirchhoff stress [Kamasamudram, Coret, and Moës 2023] by

$$\underline{\tau} = \underline{\tau}_{\text{EQ}} + \underline{\tau}_{\text{NEQ}} \quad (200)$$

with the elastic part of the Finger deformation tensor $\underline{b} = F \cdot F^\top$, and

$$\underline{\tau}_{\text{NEQ}} = 2 F \cdot \frac{\partial w_{\text{NEQ}}}{\partial C_e} \cdot F^\top = 2 \frac{\partial w_{\text{NEQ}}}{\partial \underline{b}} \cdot \underline{b} \quad (201)$$

The last equality is proved in the same way as (176)

$$\frac{\partial w_{\text{NEQ}}}{\partial \underline{b}} \cdot \underline{b} = \left(\frac{\partial w_{\text{NEQ}}}{\partial C_e} : \frac{\partial C_e}{\partial \underline{b}} \right) \cdot \underline{b} = \left(\frac{\partial w_{\text{NEQ}}}{\partial C_e} : \frac{\partial (F^\top \cdot \underline{b} \cdot F^{-1})}{\partial \underline{b}} \right) \cdot \underline{b} \quad (202)$$

$$= F \cdot \frac{\partial w_{\text{NEQ}}}{\partial C_e} \cdot F^\top \quad (203)$$

Hence, the dissipation inequality becomes

$$\underline{\tau}_{\text{NEQ}} : (F \cdot \underline{l} \cdot F^{-1}) \geq 0 \Leftrightarrow \underline{\tau}_{\text{NEQ}} : (F \cdot \underline{l} \cdot F^\top \cdot \underline{b}^{-1}) \geq 0 \quad (204)$$

Furthermore, we have

$$F \cdot \underline{l} \cdot F^\top = -\frac{1}{2} F \cdot \frac{\partial C^{-1}}{\partial t} \cdot F^\top + \frac{1}{2} (F \cdot \underline{l} \cdot F^\top - (F \cdot \underline{l} \cdot F^\top)^\top) \quad (205)$$

By definition, we have (ommiting the simple contraction dot on the following calculus to make the notations lighter)

$$\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top = \frac{1}{2} (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top + (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top)^\top) + \frac{1}{2} (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top - (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top)^\top) \quad (206)$$

$$= \frac{1}{2} (\underline{\underline{F}} \cdot \frac{\partial \underline{\underline{F}}}{\partial t} \underline{\underline{F}}^{-1} \underline{\underline{F}}^\top + \underline{\underline{F}} \cdot \underline{\underline{F}}^{-\top} \frac{\partial \underline{\underline{F}}^\top}{\partial t} \underline{\underline{F}}) + \frac{1}{2} (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top - (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top)^\top) \quad (207)$$

and

$$\frac{d}{dt} (\underline{\underline{F}} \underline{\underline{F}}^{-1}) = \frac{\partial \underline{\underline{F}}}{\partial t} \underline{\underline{F}}^{-1} + \underline{\underline{F}} \cdot \frac{\partial \underline{\underline{F}}^{-1}}{\partial t} \Leftrightarrow \frac{\partial \underline{\underline{F}}}{\partial t} \underline{\underline{F}}^{-1} = -\underline{\underline{F}} \cdot \frac{\partial \underline{\underline{F}}^{-1}}{\partial t} \quad (208)$$

$$\frac{d}{dt} (\underline{\underline{F}}^{-\top} \underline{\underline{F}}^\top) = \frac{\partial \underline{\underline{F}}^{-\top}}{\partial t} \underline{\underline{F}}^\top + \underline{\underline{F}}^{-\top} \cdot \frac{\partial \underline{\underline{F}}^\top}{\partial t} \Leftrightarrow \frac{\partial \underline{\underline{F}}^{-\top}}{\partial t} \underline{\underline{F}}^\top = -\underline{\underline{F}}^{-\top} \cdot \frac{\partial \underline{\underline{F}}^\top}{\partial t} \quad (209)$$

Then, we obtain

$$\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top = \frac{1}{2} (\underline{\underline{F}} \cdot \frac{\partial \underline{\underline{F}}}{\partial t} \underline{\underline{F}}^{-1} \underline{\underline{F}}^\top + \underline{\underline{F}} \cdot \underline{\underline{F}}^{-\top} \frac{\partial \underline{\underline{F}}^\top}{\partial t} \underline{\underline{F}}) + \frac{1}{2} (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top - (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top)^\top) \quad (210)$$

$$= -\frac{1}{2} (\underline{\underline{F}} \cdot \underline{\underline{F}} \cdot \frac{\partial \underline{\underline{F}}^{-1}}{\partial t} \underline{\underline{F}}^\top + \underline{\underline{F}} \cdot \frac{\partial \underline{\underline{F}}^{-\top}}{\partial t} \underline{\underline{F}}^\top \underline{\underline{F}}) + \frac{1}{2} (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top - (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top)^\top) \quad (211)$$

$$= -\frac{1}{2} (\underline{\underline{F}} \cdot \underline{\underline{F}} \cdot \frac{\partial \underline{\underline{F}}^{-1}}{\partial t} \underline{\underline{F}}^\top \underline{\underline{F}} + \underline{\underline{F}} \cdot \underline{\underline{F}} \underline{\underline{F}}^{-1} \cdot \frac{\partial \underline{\underline{F}}^{-\top}}{\partial t} \underline{\underline{F}}^\top \underline{\underline{F}}) + \frac{1}{2} (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top - (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top)^\top) \quad (212)$$

$$= -\frac{1}{2} \underline{\underline{F}} \cdot (\frac{\partial \underline{\underline{F}}^{-1}}{\partial t} \underline{\underline{F}}^{-\top} + \underline{\underline{F}}^{-1} \cdot \frac{\partial \underline{\underline{F}}^{-\top}}{\partial t}) \underline{\underline{F}}^\top + \frac{1}{2} (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top - (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top)^\top) \quad (213)$$

$$= -\frac{1}{2} \underline{\underline{F}} \cdot \frac{\partial \underline{\underline{C}}^{-1}}{\partial t} \underline{\underline{F}}^\top + \frac{1}{2} (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top - (\underline{\underline{F}} \cdot \underline{\underline{L}} \underline{\underline{F}}^\top)^\top) \quad (214)$$

Assume isotropy, which lets only the symmetric part of the previous development appear [Reese and Govindjee 1998]. Thus, we obtain the following dissipation inequality

$$-\underline{\underline{\tau}}_{\text{NEQ}} : (\underline{\underline{F}} \cdot \frac{\partial \underline{\underline{C}}^{-1}}{\partial t} \cdot \underline{\underline{F}}^\top \cdot \underline{\underline{b}}^{-1}) \geq 0 \quad (215)$$

which may be written in terms of the Lie derivative of the Finger elastic tensor, since

$$\mathcal{L}_v(\underline{\underline{b}}) = \underline{\underline{F}} \cdot \frac{\partial \underline{\underline{C}}^{-1}}{\partial t} \cdot \underline{\underline{F}}^\top \quad (216)$$

an expression that is proved as follows

$$\mathcal{L}_v(\underline{\underline{b}}) = \underline{\underline{F}}^\top \cdot \frac{\partial}{\partial t} (\underline{\underline{F}}^{-1} \cdot \underline{\underline{b}} \cdot \underline{\underline{F}}^{-\top}) \cdot \underline{\underline{F}} = \underline{\underline{F}}^\top \cdot \frac{\partial}{\partial t} (\underline{\underline{F}}^{-1} \cdot \underline{\underline{F}}^{-1} \cdot \underline{\underline{F}} \cdot \underline{\underline{F}}^\top \cdot \underline{\underline{F}}^{-\top} \cdot \underline{\underline{F}}^{-\top}) \cdot \underline{\underline{F}} \quad (217)$$

$$= \underline{\underline{F}} \cdot \frac{\partial}{\partial t} (\underline{\underline{F}}^{-1} \cdot \underline{\underline{F}}^{-\top}) \cdot \underline{\underline{F}} \quad (218)$$

$$= \underline{\underline{F}} \cdot \frac{\partial \underline{\underline{C}}^{-1}}{\partial t} \cdot \underline{\underline{F}}^\top \quad (219)$$

giving

$$-\underline{\underline{\tau}}_{\text{NEQ}} : (\mathcal{L}_v(\underline{\underline{b}}) \cdot \underline{\underline{b}}^{-1}) \geq 0 \quad (220)$$

A sufficient condition for this inequality to be satisfied is for the quantity to be a positive definite quadratic form, such that

$$\frac{1}{2} \mathcal{L}_v(\underline{\underline{b}}_e) \cdot \underline{\underline{b}}_e^{-1} = -\underline{\underline{V}}^{-1} : \underline{\underline{\tau}}_{\text{NEQ}} \quad (221)$$

with $\underline{\underline{V}}$ a fourth order isotropic positive definite tensor.

Then, we express this law in terms of the inverse of the inelasticity Cauchy-Green tensor. First, we know that

$$\mathcal{L}_v(\underline{\underline{b}}_e) = -2(\underline{\underline{V}}^{-1} : \underline{\underline{\tau}}_{\text{NEQ}}) \cdot \underline{\underline{b}}_e \Leftrightarrow \underline{\underline{F}} \frac{\partial \underline{\underline{C}}_i^{-1}}{\partial t} \underline{\underline{F}}^T = -4 \left[\underline{\underline{V}}^{-1} : \left(\underline{\underline{F}}_e \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_e} \cdot \underline{\underline{F}}_e^T \right) \right] \cdot \underline{\underline{b}}_e \quad (222)$$

$$\Leftrightarrow \frac{\partial \underline{\underline{C}}_i^{-1}}{\partial t} = -4 \underline{\underline{F}}^{-1} \cdot \left[\underline{\underline{V}}^{-1} : \left(\underline{\underline{F}}_e \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_e} \cdot \underline{\underline{F}}_e^T \right) \right] \cdot \underline{\underline{b}}_e \cdot \underline{\underline{F}}^{-T} \quad (223)$$

Furthermore, as we have proved (176), we obtain, noticing that $\underline{\underline{F}} = \underline{\underline{F}}^{-T} \cdot \underline{\underline{C}}$ and $\underline{\underline{b}}_e \cdot \underline{\underline{F}}^{-T} = \underline{\underline{F}} \cdot \underline{\underline{C}}_i^{-1}$

$$\frac{\partial \underline{\underline{C}}_i^{-1}}{\partial t} = -4 \underline{\underline{F}}^{-1} \left[\underline{\underline{V}}^{-1} : \left(\underline{\underline{F}} \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}} \cdot \underline{\underline{F}}^T \right) \right] \cdot \underline{\underline{b}}_e \cdot \underline{\underline{F}}^{-T} \quad (224)$$

$$= -4 \underline{\underline{F}}^{-1} \left[\underline{\underline{V}}^{-1} : \left(\underline{\underline{F}}^{-T} \cdot \underline{\underline{C}} \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}} \cdot \underline{\underline{F}}^T \right) \right] \cdot \underline{\underline{F}} \cdot \underline{\underline{C}}_i^{-1} \quad (225)$$

Then, we want to express the relationship thanks to the derivative of the out of equilibrium part of the Helmholtz free energy with respect to the inelastic part of the inelastic Cauchy Green tensor. Following [Reese and Govindjee 1997], we have

$$\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{F}}_i} = -2 \underline{\underline{C}}_e \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_e} \cdot \underline{\underline{F}}_i^{-T} \quad (226)$$

For the derivative of the Helmholtz free energy with respect to the inelastic part of the deformation gradient, one obtains

$$\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{F}}_i} = \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} : \frac{\partial \underline{\underline{C}}_i^{-1}}{\partial \underline{\underline{F}}_i} \quad (227)$$

which gives, in index notations

$$\left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \right)_{AB} \left(\frac{\partial \underline{\underline{C}}_i^{-1}}{\partial \underline{\underline{F}}_i} \right)_{APQB} \quad (228)$$

$$= \left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \right)_{AB} \frac{\partial (\underline{\underline{F}}_i^{-1} \cdot \underline{\underline{F}}_i^{-T})_{AB}}{\partial (\underline{\underline{F}}_i)_{PQ}} \quad (229)$$

$$= \left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \right)_{AB} \frac{\partial (\underline{\underline{F}}_i^{-1})_{AI} (\underline{\underline{F}}_i^{-1})_{BI}}{\partial (\underline{\underline{F}}_i)_{PQ}} \quad (230)$$

$$= - \left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \right)_{AB} (\underline{\underline{F}}_i^{-1})_{AP} (\underline{\underline{F}}_i^{-1})_{QI} (\underline{\underline{F}}_i^{-1})_{BI} - \left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \right)_{AB} (\underline{\underline{F}}_i^{-1})_{AI} (\underline{\underline{F}}_i^{-1})_{BP} (\underline{\underline{F}}_i^{-1})_{QI} \quad (231)$$

$$= - \left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \right)_{AB} (\underline{\underline{F}}_i^{-1})_{AP} (\underline{\underline{C}}_i^{-1})_{QB} - \left(\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \right)_{AB} (\underline{\underline{F}}_i^{-1})_{BP} (\underline{\underline{C}}_i^{-1})_{AQ} \quad (232)$$

$$= -(F_{\underline{\underline{i}}}^{-\top})_{PA} \left(\frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \right)_{AB} (C_{\underline{\underline{i}}}^{-1})_{BQ} - (F_{\underline{\underline{i}}}^{-\top})_{PB} \left(\left(\frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \right)^{\top} \right)_{BA} (C_{\underline{\underline{i}}}^{-1})_{AQ} \quad (233)$$

which gives

$$\frac{\partial w_{\text{NEQ}}}{\partial F_{\underline{\underline{i}}}} = \frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} : \frac{\partial C_{\underline{\underline{i}}}^{-1}}{\partial F_{\underline{\underline{i}}}} \quad (234)$$

$$= -F_{\underline{\underline{i}}}^{-\top} \cdot \left(\frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \right) \cdot C_{\underline{\underline{i}}}^{-\top} - F_{\underline{\underline{i}}}^{-\top} \cdot \left(\frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \right)^{\top} \cdot C_{\underline{\underline{i}}}^{-1} \quad (235)$$

$$= -2F_{\underline{\underline{i}}}^{-\top} \cdot \left(\frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \right)^{\top} \cdot C_{\underline{\underline{i}}}^{-1} \quad (236)$$

where we used the following symmetries

$$\left(\frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \right)^{\top} = \frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \quad (237)$$

$$C_{\underline{\underline{i}}}^{-\top} = C_{\underline{\underline{i}}}^{-1} \quad (238)$$

Then, we have

$$\frac{\partial w_{\text{NEQ}}}{\partial F_{\underline{\underline{i}}}} = -2C_{\underline{\underline{e}}} \cdot \frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{e}}}} \cdot F_{\underline{\underline{i}}}^{-\top} \quad (239)$$

$$\Leftrightarrow -2F_{\underline{\underline{i}}}^{-\top} \cdot \frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \cdot C_{\underline{\underline{i}}}^{-1} = -2C_{\underline{\underline{e}}} F_{\underline{\underline{i}}} \frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{e}}}} \quad (240)$$

$$\Leftrightarrow \frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \cdot C_{\underline{\underline{i}}}^{-1} = C_{\underline{\underline{e}}} \cdot \frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{e}}}} \quad (241)$$

finally giving the viscoelastic law in terms of the inverse of the inelastic Cauchy-Green tensor

$$\frac{\partial C_{\underline{\underline{i}}}^{-1}}{\partial t} = -4F_{\underline{\underline{i}}}^{-1} \cdot \left[V_{\underline{\underline{e}}}^{-1} : \left(F_{\underline{\underline{i}}}^{-\top} \cdot \frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \cdot C_{\underline{\underline{i}}}^{-1} \cdot F_{\underline{\underline{i}}}^{\top} \right) \right] \cdot F_{\underline{\underline{i}}} \cdot C_{\underline{\underline{i}}}^{-1} \quad (242)$$

_____ *qed.* ■

2.2. Obtaining the viscoelastic part of the dissipative bracket

Proposition 2.2

The double contraction of the derivative of an arbitrary function f with respect to the second order tensor inverse of the inelastic Cauchy Green tensor with the time derivative of the inverse inelastic Cauchy Green tensor is [Schiebl and Betsch 2021]

$$\frac{\partial f}{\partial C_{\underline{\underline{i}}}^{-1}} : \frac{\partial C_{\underline{\underline{i}}}^{-1}}{\partial t} = -4 \left[\frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \cdot C_{\underline{\underline{i}}}^{-1} \right] : \left[F_{\underline{\underline{i}}}^{-1} \cdot \left(F_{\underline{\underline{i}}}^{-1} \cdot V_{\underline{\underline{e}}}^{-1} \cdot F_{\underline{\underline{i}}} \right)^{\top} \cdot F_{\underline{\underline{i}}} \right] : \left[\frac{\partial f}{\partial C_{\underline{\underline{i}}}^{-1}} \cdot C_{\underline{\underline{i}}}^{-1} \right] \quad (243)$$

Proof _____

We can start from the following equality

$$\frac{\partial f}{\partial C_{\underline{\underline{i}}}^{-1}} : \frac{\partial C_{\underline{\underline{i}}}^{-1}}{\partial t} = -4 \frac{\partial f}{\partial C_{\underline{\underline{i}}}^{-1}} : \left[F_{\underline{\underline{i}}}^{-1} \cdot \left[V_{\underline{\underline{e}}}^{-1} : \left(F_{\underline{\underline{i}}}^{-\top} \cdot \frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \cdot C_{\underline{\underline{i}}}^{-1} \cdot F_{\underline{\underline{i}}}^{\top} \right) \right] \cdot F_{\underline{\underline{i}}} \cdot C_{\underline{\underline{i}}}^{-1} \right] \quad (244)$$

The inelastic flow tensor is purely Eulerian $[V^{-1}]_{abcd}$. We will make use of the following tensor relationships [Itskov 2019]

$$\left[\underline{\underline{X}} : \underline{\underline{A}} \right]_{jk} = A_{ijkl} X_{il} \quad (245)$$

$$\left[\underline{\underline{A}} : \underline{\underline{X}} \right]_{il} = A_{ijkl} X_{jk} \quad (246)$$

$$\left[\underline{\underline{X}} \cdot \underline{\underline{A}} \right]_{ijkl} = X_{ip} A_{pjkl} := \left[\underline{\underline{X}} \underline{\underline{A}} \right]_{ijkl} \quad (247)$$

$$\left[\underline{\underline{A}} \cdot \underline{\underline{X}} \right]_{ijkl} = A_{ijkp} X_{pl} := \left[\underline{\underline{A}} \underline{\underline{X}} \right]_{ijkl} \quad (248)$$

$$\left[\underline{\underline{X}} \cdot \underline{\underline{Y}} \right]_{il} = X_{ip} Y_{pl} := \left[\underline{\underline{X}} \underline{\underline{Y}} \right] \quad (249)$$

Following this, the calculus yields

$$\frac{\partial f}{\partial \underline{\underline{C}}_i^{-1}} : \frac{\partial \underline{\underline{C}}_i^{-1}}{\partial t} = -4 \frac{\partial f}{\partial \underline{\underline{C}}_i^{-1}} : \left[\underline{\underline{F}}^{-1} \cdot \left(\underline{\underline{V}}^{-1} : \left(\underline{\underline{F}}^{-\top} \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \cdot \underline{\underline{C}}_i^{-1} \cdot \underline{\underline{F}}^\top \right) \right) \cdot \underline{\underline{F}} \cdot \underline{\underline{C}}_i^{-1} \right] \quad (250)$$

$$= -4 \left[\frac{\partial f}{\partial \underline{\underline{C}}_i^{-1}} \right]_{AB} \left[\underline{\underline{F}}^{-1} \cdot \left(\underline{\underline{V}}^{-1} : \left(\underline{\underline{F}}^{-\top} \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \cdot \underline{\underline{C}}_i^{-1} \cdot \underline{\underline{F}}^\top \right) \right) \cdot \underline{\underline{F}} \cdot \underline{\underline{C}}_i^{-1} \right]_{AB} \quad (251)$$

$$= -4 \left[\frac{\partial f}{\partial \underline{\underline{C}}_i^{-1}} \right]_{AB} [F^{-1}]_{Aa} \left[\underline{\underline{V}}^{-1} : \left(\underline{\underline{F}}^{-\top} \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \cdot \underline{\underline{C}}_i^{-1} \cdot \underline{\underline{F}}^\top \right) \right]_{ab} \left[\underline{\underline{F}} \cdot \underline{\underline{C}}_i^{-1} \right]_{bB} \quad (252)$$

$$= -4 \left[\frac{\partial f}{\partial \underline{\underline{C}}_i^{-1}} \right]_{AB} [F^{-1}]_{Aa} [V^{-1}]_{apqb} \left[\underline{\underline{F}}^{-\top} \cdot \frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \cdot \underline{\underline{C}}_i^{-1} \cdot \underline{\underline{F}}^\top \right]_{pq} [F]_{bc} [C_i^{-1}]_{CB} \quad (253)$$

$$= -4 \left[\frac{\partial f}{\partial \underline{\underline{C}}_i^{-1}} \cdot \underline{\underline{C}}_i^{-\top} \right]_{AC} [F^{-1}]_{Aa} [V^{-1}]_{apqb} [F^{-\top}]_{pQ} \left[\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \cdot \underline{\underline{C}}_i^{-1} \right]_{QR} [F^\top]_{Rq} [F]_{bc} \quad (254)$$

$$= -4 \left[\frac{\partial f}{\partial \underline{\underline{C}}_i^{-1}} \cdot \underline{\underline{C}}_i^{-1} \right]_{AC} [F^{-\top}]_{pQ} [F^{-1}]_{Aa} [V^{-1}]_{apqb} [F]_{bc} [F^\top]_{Rq} \left[\frac{\partial w_{\text{NEQ}}}{\partial \underline{\underline{C}}_i^{-1}} \cdot \underline{\underline{C}}_i^{-1} \right]_{QR} \quad (255)$$

employing the symmetry of the inverse of the Cauchy-Green inelastic tensor. Furthermore, we have

$$[F^{-\top}]_{pQ} [F^{-1}]_{Aa} [V^{-1}]_{apqb} [F]_{bc} [F^\top]_{Rq} = [F^{-\top}]_{pQ} \left[\underline{\underline{F}}^{-1} \cdot \underline{\underline{V}}^{-1} \cdot \underline{\underline{F}} \right]_{ApqC} [F^\top]_{Rq} \quad (256)$$

$$= [F^{-\top}]_{pQ} \left[\left(\underline{\underline{F}}^{-1} \cdot \underline{\underline{V}}^{-1} \cdot \underline{\underline{F}} \right)^\top \right]_{pACq} [F^\top]_{Rq} \quad (257)$$

$$= \left[\underline{\underline{F}}^{-1} \cdot \left(\underline{\underline{F}}^{-1} \cdot \underline{\underline{V}}^{-1} \cdot \underline{\underline{F}} \right)^\top \cdot \underline{\underline{F}} \right]_{QACR} \quad (258)$$

applying definition of the transposition

$$[A^\top]_{abcd} = [A]_{badc} \quad (259)$$

Hence, we have

$$\frac{\partial f}{\partial C_{\underline{\underline{i}}}^{-1}} : \frac{\partial C_{\underline{\underline{i}}}^{-1}}{\partial t} = -4 \left[\frac{\partial f}{\partial C_{\underline{\underline{i}}}^{-1}} \cdot C_{\underline{\underline{i}}}^{-1} \right]_{AC} \left[F^{-1} \cdot \left(F^{-1} \cdot V^{-1} \cdot F \right)^{\top} \cdot F \right]_{QACR} \left[\frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \cdot C_{\underline{\underline{i}}}^{-1} \right]_{QR} \quad (260)$$

$$\Leftrightarrow \frac{\partial f}{\partial C_{\underline{\underline{i}}}^{-1}} : \frac{\partial C_{\underline{\underline{i}}}^{-1}}{\partial t} = -4 \left[\frac{\partial w_{\text{NEQ}}}{\partial C_{\underline{\underline{i}}}^{-1}} \cdot C_{\underline{\underline{i}}}^{-1} \right] : \left[F^{-1} \cdot \left(F^{-1} \cdot V^{-1} \cdot F \right)^{\top} \cdot F \right] : \left[\frac{\partial f}{\partial C_{\underline{\underline{i}}}^{-1}} \cdot C_{\underline{\underline{i}}}^{-1} \right] \quad (261)$$

_____ *qed.* ■

2.3. Symmetry of the fourth order tensor within the viscous dissipative bracket

Proposition 2.3

The Lagrangian fourth order tensor of inelastic flow [Schiebl and Betsch 2021] is symmetric

$$\left[F^{-1} \cdot \left(F^{-1} \cdot V^{-1} \cdot F \right)^{\top} \cdot F \right] = \left[F^{-1} \cdot \left(F^{-1} \cdot V^{-1} \cdot F \right)^{\top} \cdot F \right]^{\top} \quad (262)$$

where

$$\left[A^{\top} \right]_{abcd} = [A]_{badc} \quad (263)$$

Proof _____

We have

$$\left[F^{-1} \cdot \left(F^{-1} \cdot V^{-1} \cdot F \right)^{\top} \cdot F \right]_{QACR} = \left[F^{-\top} \right]_{pQ} [F^{-1}]_{Aa} [V^{-1}]_{apqb} [F]_{bc} [F^{\top}]_{Rq} \quad (264)$$

$$= \left[F^{-\top} \right]_{pQ} [F^{-1}]_{Aa} [V^{-\top}]_{pabq} [F]_{bc} [F^{\top}]_{Rq} \quad (265)$$

$$= [F^{-1}]_{Aa} \left[\left(F^{-1} \cdot V^{-\top} \cdot F \right) \right]_{QabR} [F]_{bc} \quad (266)$$

$$= [F^{-1}]_{Aa} \left[\left(F^{-1} \cdot V^{-\top} \cdot F \right)^{\top} \right]_{aQRb} [F]_{bc} \quad (267)$$

$$= \left[F^{-1} \cdot \left(F^{-1} \cdot V^{-\top} \cdot F \right)^{\top} \cdot F \right]_{AQRC} \quad (268)$$

Since V^{-1} , is a fourth order isotropic tensor, it can be written as [Reese and Govindjee 1998]

$$V^{-1} = \frac{1}{2\eta_D} \left(I - \frac{1}{3} I \otimes I \right) + \frac{1}{9\eta_V} I \otimes I \quad (269)$$

where η_D and η_V represent the deviatoric and volumetric viscosities. The fourth order identity tensor has only the symmetry property $I = I^{\top}$. Thus, the flow tensor has the property

$$V^{-1} = V^{-\top} \quad (270)$$

thus proving the symmetry of

$$\underline{\underline{N}} = \left[\underline{\underline{F}}^{-1} \cdot \left(\underline{\underline{F}}^{-1} \cdot \underline{\underline{V}}^{-1} \cdot \underline{\underline{F}} \right)^{\top} \cdot \underline{\underline{F}} \right] \quad (271)$$

_____ *qed.* ■

2.4. Conduction bracket

Proposition 2.4

The integral within the Double Bracket Structure concerning the conduction evolution of temperature yields

$$\int_{\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} \left(-\frac{1}{c} \text{DIV}_X Q \right) d\omega_X \quad (272)$$

$$= \int_{\omega_X} \frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) \cdot \underline{\underline{K}} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial X} \left(\frac{\partial s^*}{\partial T} \right) d\omega_X - \int_{\partial\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} \underline{\underline{K}} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial X} \left(\frac{\partial s^*}{\partial T} \right) \cdot \underline{\underline{N}} d(\partial\omega_X) \quad (273)$$

Proof _____

First, the conduction term within the evolution of temperature yields

$$-\frac{1}{c} \text{DIV}_X Q = \frac{1}{c} \frac{\partial}{\partial X} \left(\underline{\underline{K}} \frac{\partial T}{\partial X} \right) = \frac{1}{c} \frac{\partial}{\partial X} \left(\underline{\underline{K}} \frac{\partial}{\partial X} \left(c \left(\frac{\partial s^*}{\partial T} \right)^{-1} \right) \right) \quad (274)$$

$$= -\frac{\partial}{\partial X} \left(\underline{\underline{K}} \frac{1}{\left(\frac{\partial s^*}{\partial T} \right)^2} \frac{\partial}{\partial X} \left(\frac{\partial s^*}{\partial T} \right) \right) = -\frac{\partial}{\partial X} \left(\underline{\underline{K}} \left(\frac{T}{c} \right)^2 \frac{\partial}{\partial X} \left(\frac{\partial s^*}{\partial T} \right) \right) \quad (275)$$

Then, we have

$$\int_{\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} \left(-\frac{1}{c} \text{DIV}_X Q \right) d\omega_X = \int_{\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} \left(-\frac{\partial}{\partial X} \left(\underline{\underline{K}} \frac{1}{\left(\frac{\partial s^*}{\partial T} \right)^2} \frac{\partial}{\partial X} \left(\frac{\partial s^*}{\partial T} \right) \right) \right) d\omega_X \quad (276)$$

$$= \int_{\omega_X} -\frac{1}{\rho} \frac{\partial f}{\partial T} \frac{\partial}{\partial X} \left(\underline{\underline{K}} \left(\frac{T}{c} \right)^2 \frac{\partial}{\partial X} \left(\frac{\partial s^*}{\partial T} \right) \right) d\omega_X \quad (277)$$

$$= \int_{\omega_X} \frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) \cdot \underline{\underline{K}} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial X} \left(\frac{\partial s^*}{\partial T} \right) d\omega_X - \int_{\partial\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} \underline{\underline{K}} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial X} \left(\frac{\partial s^*}{\partial T} \right) \cdot \underline{\underline{N}} d(\partial\omega_X) \quad (278)$$

_____ *qed.* ■

2.5. Skew-symmetry of the reversible bracket

Proposition 2.5

The reversible bracket defined in (163) is skew-symmetric.

Proof _____

$$\{\mathcal{F}, \mathcal{G}\} + \{\mathcal{G}, \mathcal{F}\} \quad (279)$$

$$= \int_{\omega_X} \left(\frac{\partial f}{\partial \underline{\underline{\chi}}} \cdot \frac{\partial g}{\partial \underline{\underline{p}}} - \frac{\partial f}{\partial \underline{\underline{p}}} \cdot \frac{\partial g}{\partial \underline{\underline{\chi}}} - \frac{\partial f}{\partial \underline{\underline{\Pi}}} : \underline{\underline{\nabla}}_X \frac{\partial g}{\partial \underline{\underline{p}}} S^+ + \underline{\underline{\nabla}}_X \frac{\partial f}{\partial \underline{\underline{p}}} : \frac{\partial g}{\partial \underline{\underline{\Pi}}} S^+ \right. \quad (280)$$

$$\left. + \frac{\partial f}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} : \underline{\underline{\nabla}}_X \frac{\partial g}{\partial \underline{\underline{p}}} S^- - \frac{\partial g}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \underline{\underline{\nabla}}_X \frac{\partial f}{\partial \underline{\underline{p}}} : \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} S^+ \right) \quad (281)$$

$$+ \frac{\partial g}{\partial \underline{\underline{\chi}}} \cdot \frac{\partial f}{\partial \underline{\underline{p}}} - \frac{\partial g}{\partial \underline{\underline{p}}} \cdot \frac{\partial f}{\partial \underline{\underline{\chi}}} - \frac{\partial g}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial f}{\partial \underline{\underline{p}}} \underline{\underline{S}} + \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} : \frac{\partial f}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} + \quad (282)$$

$$+ \frac{\partial g}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial f}{\partial \underline{\underline{p}}} \underline{\underline{S}} - \frac{\partial f}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} : \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} \Big) d\omega_X \quad (283)$$

$$= \int_{\omega_X} \left(\frac{\partial f}{\partial \underline{\underline{\chi}}} \cdot \frac{\partial g}{\partial \underline{\underline{p}}} - \frac{\partial f}{\partial \underline{\underline{p}}} \cdot \frac{\partial g}{\partial \underline{\underline{\chi}}} + \frac{\partial g}{\partial \underline{\underline{\chi}}} \cdot \frac{\partial f}{\partial \underline{\underline{p}}} - \frac{\partial g}{\partial \underline{\underline{p}}} \cdot \frac{\partial f}{\partial \underline{\underline{\chi}}} \right. \quad (284)$$

$$\left. - \frac{\partial f}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} \underline{\underline{S}} + \nabla_{\underline{\underline{x}}} \frac{\partial f}{\partial \underline{\underline{p}}} : \frac{\partial g}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} - \frac{\partial g}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial f}{\partial \underline{\underline{p}}} \underline{\underline{S}} + \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} : \frac{\partial f}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} \right. \quad (285)$$

$$\left. + \frac{\partial f}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} \underline{\underline{S}} - \frac{\partial g}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \nabla_{\underline{\underline{x}}} \frac{\partial f}{\partial \underline{\underline{p}}} : \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} + \right. \quad (286)$$

$$\left. + \frac{\partial g}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial f}{\partial \underline{\underline{p}}} \underline{\underline{S}} - \frac{\partial f}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} : \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} \right) d\omega_X \quad (287)$$

$$= \int_{\omega_X} \left(- \frac{\partial f}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} \underline{\underline{S}} + \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} : \frac{\partial f}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} + \nabla_{\underline{\underline{x}}} \frac{\partial f}{\partial \underline{\underline{p}}} : \frac{\partial g}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} - \frac{\partial g}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial f}{\partial \underline{\underline{p}}} \underline{\underline{S}} \right. \quad (288)$$

$$\left. + \frac{\partial f}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} \underline{\underline{S}} - \frac{\partial f}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} : \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} + \right. \quad (289)$$

$$\left. - \frac{\partial g}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \nabla_{\underline{\underline{x}}} \frac{\partial f}{\partial \underline{\underline{p}}} : \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} + \frac{\partial g}{\partial T} \left(\frac{\partial s^*}{\partial T} \right)^{-1} \frac{\partial s^*}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial f}{\partial \underline{\underline{p}}} \underline{\underline{S}} \right) d\omega_X \quad (290)$$

Every term in the last equation is computed using the symmetry of the second Piola Kirchhoff stress tensor (and of the double contraction operation). For instance, taking the first term leads to

$$- \frac{\partial f}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} \underline{\underline{S}} + \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} : \frac{\partial f}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} = - \frac{\partial f}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} \underline{\underline{S}} + \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} \underline{\underline{S}}^T : \frac{\partial f}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} \quad (291)$$

$$= - \frac{\partial f}{\partial \underline{\underline{\Pi}}} : \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} \underline{\underline{S}} + \nabla_{\underline{\underline{x}}} \frac{\partial g}{\partial \underline{\underline{p}}} \underline{\underline{S}} : \frac{\partial f}{\partial \underline{\underline{\Pi}}} \underline{\underline{S}} = 0 \quad (292)$$

Using the same calculus on each of the terms finally leads to

$$\{\mathcal{F}, \mathcal{G}\} + \{\mathcal{G}, \mathcal{F}\} = 0 \quad \forall \mathcal{F}, \mathcal{G} \quad (293)$$

qed. ■

2.6. Symmetry of the dissipative bracket

Proposition 2.6

The dissipative bracket defined in (164) is symmetric.

Proof _____

$$(\mathcal{F}, \mathcal{G}) - (\mathcal{G}, \mathcal{F}) \quad (294)$$

$$= \int_{\omega_X} \left(\frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) \cdot \underline{\underline{K}} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial X} \left(\frac{\partial g}{\partial T} \right) + \frac{\partial f}{\partial T} \frac{4}{c^2} \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\text{ii}}^{-1}} \cdot C_{\text{ii}}^{-1} \right) : T \underline{\underline{N}} : \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\text{ii}}^{-1}} \cdot C_{\text{ii}}^{-1} \right) \frac{\partial g}{\partial T} + \right. \quad (295)$$

$$\left. - \frac{\partial f}{\partial T} \frac{4}{c} \left(\frac{\partial g}{\partial C_{\text{ii}}^{-1}} \cdot C_{\text{ii}}^{-1} \right) : T \underline{\underline{N}} : \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\text{ii}}^{-1}} \cdot C_{\text{ii}}^{-1} \right) - \frac{\partial g}{\partial T} \frac{4}{c} \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\text{ii}}^{-1}} \cdot C_{\text{ii}}^{-1} \right) : T \underline{\underline{N}} : \left(\frac{\partial f}{\partial C_{\text{ii}}^{-1}} \cdot C_{\text{ii}}^{-1} \right) + \right. \quad (296)$$

$$+ 4 \left(\frac{\partial g}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial f}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) d\omega_X + \quad (297)$$

$$+ \int_{\omega_X} - \left(\frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial g}{\partial T} \right) \cdot K \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) - \frac{\partial g}{\partial T} \frac{4}{c^2} \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) \frac{\partial f}{\partial T} + \quad (298)$$

$$+ \frac{\partial g}{\partial T} \frac{4}{c} \left(\frac{\partial f}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) + \frac{\partial f}{\partial T} \frac{4}{c} \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial g}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) + \quad (299)$$

$$- 4 \left(\frac{\partial f}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial g}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) d\omega_X \quad (300)$$

Due to the symmetry of the conduction operator $\underline{\underline{K}}$ and the lagrangian viscous fourth order tensor $\underline{\underline{N}}$, we have

$$\underline{a} \cdot \underline{\underline{K}} \cdot \underline{b} = \underline{b} \cdot \underline{\underline{K}}^T \cdot \underline{a} = \underline{b} \cdot \underline{\underline{K}} \cdot \underline{a} \quad \forall \underline{a}, \underline{b} \quad (301)$$

$$\underline{\underline{X}} : \underline{\underline{N}} : \underline{\underline{Y}} = \underline{\underline{Y}} : \underline{\underline{N}}^T : \underline{\underline{X}} = \underline{\underline{Y}} : \underline{\underline{N}} : \underline{\underline{X}} \quad \forall \underline{\underline{X}}, \underline{\underline{Y}} \quad (302)$$

Then, we obtain

$$(\mathcal{F}, \mathcal{G}) - (\mathcal{G}, \mathcal{F}) \quad (303)$$

$$= \int_{\omega_X} \underbrace{\left(\frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) \cdot \underline{\underline{K}} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial X} \left(\frac{\partial g}{\partial T} \right) + \frac{\partial f}{\partial T} \frac{4}{c^2} \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) \frac{\partial g}{\partial T} +}_{=(a)} \quad (304)$$

$$\underbrace{- \frac{\partial f}{\partial T} \frac{4}{c} \left(\frac{\partial g}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right)}_{=-(c)} - \underbrace{\frac{\partial g}{\partial T} \frac{4}{c} \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial f}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right)}_{=-(d)} + \quad (305)$$

$$+ 4 \left(\frac{\partial g}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial f}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) d\omega_X + \quad (306)$$

$$+ \int_{\omega_X} \underbrace{\left(- \frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) \cdot \underline{\underline{K}} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial X} \left(\frac{\partial g}{\partial T} \right) - \frac{\partial g}{\partial T} \frac{4}{c^2} \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) \frac{\partial f}{\partial T} +}_{=-(a)} \quad (307)$$

$$+ \underbrace{\frac{\partial g}{\partial T} \frac{4}{c} \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial f}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right)}_{=(d)} + \underbrace{\frac{\partial f}{\partial T} \frac{4}{c} \left(\frac{\partial g}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right)}_{=(c)} + \quad (308)$$

$$\underbrace{- 4 \left(\frac{\partial g}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) : T N : \left(\frac{\partial f}{\partial C_{\Xi}^{-1}} \cdot C_{\Xi}^{-1} \right) d\omega_X}_{=-(e)} \quad (309)$$

$$= 0 \quad (310)$$

qed. ■

2.7. Positivity of the dissipative bracket

Proposition 2.7

The dissipative bracket defined in (164) is positive.

Proof _____

$$(\mathcal{F}, \mathcal{F}) = \int_{\omega_X} \left[\frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) \cdot \underline{\underline{K}} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) + \right. \quad (311)$$

$$\left. + \frac{\partial f}{\partial T} \frac{4}{c^2} \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C^{-1}} \cdot C^{-1} \right) : T \underline{\underline{N}} : \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C^{-1}} \cdot C^{-1} \right) \frac{\partial f}{\partial T} + \right. \quad (312)$$

$$\left. - \frac{\partial f}{\partial T} \frac{4}{c} \left(\frac{\partial f}{\partial C^{-1}} \cdot C^{-1} \right) : T \underline{\underline{N}} : \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C^{-1}} \cdot C^{-1} \right) - \frac{\partial f}{\partial T} \frac{4}{c} \left(\frac{\partial \hat{e}_{\text{int}}^*}{\partial C^{-1}} \cdot C^{-1} \right) : T \underline{\underline{N}} : \left(\frac{\partial f}{\partial C^{-1}} \cdot C^{-1} \right) + \right. \quad (313)$$

$$\left. + 4 \left(\frac{\partial f}{\partial C^{-1}} \cdot C^{-1} \right) : T \underline{\underline{N}} : \left(\frac{\partial f}{\partial C^{-1}} \cdot C^{-1} \right) \right] d\omega_X \quad (314)$$

$$= \int_{\omega_X} \left[\frac{1}{\rho} \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) \cdot \underline{\underline{K}} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial X} \left(\frac{\partial f}{\partial T} \right) + \right. \quad (315)$$

$$\left. + \left(\frac{\partial f}{\partial T} \frac{2}{c} \frac{\partial \hat{e}_{\text{int}}^*}{\partial C^{-1}} \cdot C^{-1} - 2 \frac{\partial f}{\partial C^{-1}} \cdot C^{-1} \right) : T \underline{\underline{N}} : \left(\frac{\partial f}{\partial T} \frac{2}{c} \frac{\partial \hat{e}_{\text{int}}^*}{\partial C^{-1}} \cdot C^{-1} - 2 \frac{\partial f}{\partial C^{-1}} \cdot C^{-1} \right) \right] d\omega_X \quad (316)$$

The density, as well as the temperature, are positive. Moreover, $\underline{\underline{K}}$ and $\underline{\underline{N}}$ are positive tensors. Therefore, we ensure the positivity of the dissipative bracket.

_____ *qed.* ■

2.8. The Jacobi identity

We adopt the following notation to make our notations lighter

$$\{\mathcal{F}, \mathcal{G}\} = \int_{\omega_X} \left[f_{,\underline{\underline{\chi}}} \cdot g_{,\underline{\underline{p}}} - f_{,\underline{\underline{p}}} \cdot g_{,\underline{\underline{\chi}}} + f_{,\underline{\underline{pX}}} : g_{,\underline{\underline{\Pi}}} \underline{\underline{S}} - f_{,\underline{\underline{\Pi}}} : g_{,\underline{\underline{pX}}} \underline{\underline{S}} + \right. \quad (317)$$

$$\left. - g_{,T} (s_{,T}^*)^{-1} f_{,\underline{\underline{pX}}} : s_{,\underline{\underline{\Pi}}}^* \underline{\underline{S}} + f_{,T} (s_{,T}^*)^{-1} s_{,\underline{\underline{\Pi}}}^* : g_{,\underline{\underline{pX}}} \underline{\underline{S}} \right] d\omega_X \quad (318)$$

We must verify that the reversible bracket is consistent with the Jacobi identity

$$\{\{\mathcal{F}, \mathcal{G}\}, \mathcal{H}\} + \{\{\mathcal{G}, \mathcal{H}\}, \mathcal{F}\} + \{\{\mathcal{H}, \mathcal{F}\}, \mathcal{G}\} = 0 \quad \forall \mathcal{F}, \mathcal{G}, \mathcal{H} \quad (319)$$

We consider physical assumptions on the derivatives off the densities f, g, h , based on the type of hypothesis that we verify on the energy. In general, the total energy would be stated as

$$\hat{\mathcal{E}}_{\text{tot}} = \int_{\omega_X} \hat{E}_{\text{tot}} d\omega_X = \int_{\omega_X} \left(\frac{1}{2} \frac{1}{\rho} \underline{\underline{p}} \cdot \underline{\underline{p}} + w^*(\underline{\underline{\Pi}}, T) - \underline{\underline{F}}^{\text{ext}} \cdot \underline{\underline{\chi}} \right) d\omega_X \quad (320)$$

The double derivatives of the energy involving cross terms are equal to zero. Moreover, see that the space derivative of the double derivative of $\hat{E}_{\text{tot}}$ with respect to the linear momentum is equal to zero, as it is the derivative of a constant. These observations are translated into conditions on the function f (and therefore g and h , present within the Jacobi identity).

$$\frac{\partial^2 f}{\partial \underline{\underline{\chi}} \partial \underline{\underline{p}}} = \frac{\partial^2 f}{\partial \underline{\underline{p}} \partial \underline{\underline{\chi}}} = 0 \quad (321)$$

$$\frac{\partial^2 f}{\partial \underline{\chi} \partial \underline{\Pi}} = \frac{\partial^2 f}{\partial \underline{\Pi} \partial \underline{\chi}} = \underline{\underline{0}} \quad (322)$$

$$\frac{\partial^2 f}{\partial \underline{p} \partial \underline{\Pi}} = \frac{\partial^2 f}{\partial \underline{\Pi} \partial \underline{p}} = \underline{\underline{0}} \quad (323)$$

$$\frac{\partial^2 f}{\partial T \partial \underline{p}} = \frac{\partial^2 f}{\partial \underline{p} \partial T} = \underline{\underline{0}} \quad (324)$$

$$\frac{\partial^2 f}{\partial T \partial \underline{\chi}} = \frac{\partial^2 f}{\partial \underline{\chi} \partial T} = \underline{\underline{0}} \quad (325)$$

$$\frac{\partial^3 f}{\partial \underline{X} \partial \underline{p} \partial \underline{p}} = \frac{\partial}{\partial \underline{X}} \left(\frac{\partial^2 f}{\partial \underline{p} \partial \underline{p}} \right) = \underline{\underline{0}} \quad (326)$$

Furthermore, consider the following assumptions on the second Piola Kirchhoff stress tensor

$$\frac{\partial \underline{\underline{S}}}{\partial \underline{\chi}} = \underline{\underline{0}}, \quad \frac{\partial \underline{\underline{S}}}{\partial \underline{p}} = \underline{\underline{0}}, \quad \frac{\partial \underline{\underline{S}}}{\partial T} = \underline{\underline{0}} \quad (327)$$

Finally, we make the following assumption on the entropy, consistent with the assumptions made in the small strain setting

$$\frac{\partial s^*}{\partial \underline{\chi}} = \underline{\underline{0}} \quad \frac{\partial s^*}{\partial \underline{p}} = \underline{\underline{0}} \quad (328)$$

Proof

We develop each of the terms of the the Jacobi identity, and identify each of the term with its opposite within the remaining brackets

$$\{\{\mathcal{F}, \mathcal{G}\}, \mathcal{H}\} \quad (329)$$

$$= \left\{ \int_{\omega_X} \left[f_{,\underline{\chi}} \cdot g_{,p} - f_{,p} \cdot g_{,\underline{\chi}} + f_{,pX} : g_{,\underline{\Pi}\underline{\underline{S}}} - f_{,\underline{\Pi}} : g_{,pX\underline{\underline{S}}} - g_{,T(s_T^*)}^{-1} f_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* + f_{,T(s_T^*)}^{-1} s_{,\underline{\Pi}}^* : g_{,pX\underline{\underline{S}}} \right] d\omega_X, \mathcal{H} \right\} \quad (330)$$

$$= \int_{\omega_X} \left\{ \left[f_{,\underline{\chi}} \cdot g_{,p} - f_{,p} \cdot g_{,\underline{\chi}} + f_{,pX} : g_{,\underline{\Pi}\underline{\underline{S}}} - f_{,\underline{\Pi}} : g_{,pX\underline{\underline{S}}} - g_{,T(s_T^*)}^{-1} f_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* + f_{,T(s_T^*)}^{-1} s_{,\underline{\Pi}}^* : g_{,pX\underline{\underline{S}}} \right]_{,\underline{\chi}} \cdot h_{,p} + \right. \quad (331)$$

$$\left. - \left[f_{,\underline{\chi}} \cdot g_{,p} - f_{,p} \cdot g_{,\underline{\chi}} + f_{,pX} : g_{,\underline{\Pi}\underline{\underline{S}}} - f_{,\underline{\Pi}} : g_{,pX\underline{\underline{S}}} - g_{,T(s_T^*)}^{-1} f_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* + f_{,T(s_T^*)}^{-1} s_{,\underline{\Pi}}^* : g_{,pX\underline{\underline{S}}} \right]_{,p} \cdot h_{,\underline{\chi}} + \right. \quad (332)$$

$$\left. + \left[f_{,\underline{\chi}} \cdot g_{,p} - f_{,p} \cdot g_{,\underline{\chi}} + f_{,pX} : g_{,\underline{\Pi}\underline{\underline{S}}} - f_{,\underline{\Pi}} : g_{,pX\underline{\underline{S}}} - g_{,T(s_T^*)}^{-1} f_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* + f_{,T(s_T^*)}^{-1} s_{,\underline{\Pi}}^* : g_{,pX\underline{\underline{S}}} \right]_{,pX} : h_{,\underline{\Pi}\underline{\underline{S}}} + \right. \quad (333)$$

$$\left. - \left[f_{,\underline{\chi}} \cdot g_{,p} - f_{,p} \cdot g_{,\underline{\chi}} + f_{,pX} : g_{,\underline{\Pi}\underline{\underline{S}}} - f_{,\underline{\Pi}} : g_{,pX\underline{\underline{S}}} - g_{,T(s_T^*)}^{-1} f_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* + f_{,T(s_T^*)}^{-1} s_{,\underline{\Pi}}^* : g_{,pX\underline{\underline{S}}} \right]_{,\underline{\Pi}} : h_{,pX\underline{\underline{S}}} + \right. \quad (334)$$

$$\left. - h_{,T(s_T^*)}^{-1} \left[f_{,\underline{\chi}} \cdot g_{,p} - f_{,p} \cdot g_{,\underline{\chi}} + f_{,pX} : g_{,\underline{\Pi}\underline{\underline{S}}} - f_{,\underline{\Pi}} : g_{,pX\underline{\underline{S}}} + \right. \right. \quad (335)$$

$$\left. - g_{,T(s_T^*)}^{-1} f_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* + f_{,T(s_T^*)}^{-1} s_{,\underline{\Pi}}^* : g_{,pX\underline{\underline{S}}} \right]_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* + \right. \quad (336)$$

$$\left. + \left[f_{,\underline{\chi}} \cdot g_{,p} - f_{,p} \cdot g_{,\underline{\chi}} + f_{,pX} : g_{,\underline{\Pi}\underline{\underline{S}}} - f_{,\underline{\Pi}} : g_{,pX\underline{\underline{S}}} + \right. \right. \quad (337)$$

$$\left. - g_{,T(s_T^*)}^{-1} f_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* + f_{,T(s_T^*)}^{-1} s_{,\underline{\Pi}}^* : g_{,pX\underline{\underline{S}}} \right]_{,T(s_T^*)}^{-1} s_{,\underline{\Pi}}^* : h_{,pX\underline{\underline{S}}} \Big\} d\omega_X \quad (338)$$

$$= \int_{\omega_X} \left\{ \underbrace{f_{,\underline{\chi}\underline{\chi}} \cdot g_{,p} \cdot h_{,p} - f_{,p} \cdot g_{,\underline{\chi}\underline{\chi}} \cdot h_{,p}}_{(1)} - \underbrace{f_{,\underline{\chi}} \cdot g_{,pp} \cdot h_{,\underline{\chi}} + f_{,pp} \cdot g_{,\underline{\chi}} \cdot h_{,\underline{\chi}}}_{(2)} + \right. \quad (339)$$

$$\underbrace{- f_{,pX} : g_{,\underline{\Pi}\underline{\underline{S}}} : h_{,pX\underline{\underline{S}}} - f_{,pX} : g_{,\underline{\Pi}\underline{\underline{S}}} : h_{,pX\underline{\underline{S}}} + f_{,\underline{\Pi}\underline{\underline{S}}} : g_{,pX\underline{\underline{S}}} : h_{,pX\underline{\underline{S}}} + f_{,\underline{\Pi}} : g_{,pX\underline{\underline{S}}} : h_{,pX\underline{\underline{S}}} +}_{(5)} \quad (340)$$

$$\underbrace{+ g_{,T(s_T^*)}^{-1} f_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* : h_{,pX\underline{\underline{S}}} + g_{,T(s_T^*)}^{-1} f_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* : h_{,pX\underline{\underline{S}}} + g_{,T(s_T^*)}^{-1} f_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* : h_{,pX\underline{\underline{S}}} +}_{(9)} \quad (341)$$

$$\underbrace{+g_{,T}(s_T^*)^{-1}f_{,pX}:s_{\Pi,\Pi}^*S:h_{,pX}S-f_{,T\Pi}(s_T^*)^{-1}s_{\Pi}^*:g_{,pX}S:h_{,pX}S}_{(12)}-\underbrace{f_{,T}(s_T^*)^{-1}s_{\Pi}^*:g_{,pX}S:h_{,pX}S}_{(13)}-\underbrace{f_{,T}(s_T^*)^{-1}s_{\Pi}^*:g_{,pX}S:h_{,pX}S}_{(14)}+ \quad (342)$$

$$\underbrace{-f_{,T}(s_T^*)^{-1}s_{\Pi\Pi}^*:g_{,pX}S:h_{,pX}S}_{(15)}-\underbrace{f_{,T}(s_T^*)^{-1}s_{\Pi}^*:g_{,pX}S:h_{,pX}S}_{(16)}+\underbrace{f_{,pX}:g_{,\Pi T}S(s_T^*)^{-1}s_{\Pi}^*:h_{,pX}S}_{(17)}+ \quad (343)$$

$$\underbrace{-f_{,\Pi T}:g_{,pX}S(s_T^*)^{-1}s_{\Pi}^*:h_{,pX}S}_{(18)}-\underbrace{g_{,TT}(s_T^*)^{-1}f_{,pX}:s_{\Pi}^*S:h_{,pX}S}_{(19)}-\underbrace{g_{,T}(s_T^*)^{-1}f_{,pX}:s_{\Pi}^*S:h_{,pX}S}_{(20)}+ \quad (344)$$

$$\underbrace{-g_{,T}(s_T^*)^{-1}f_{,pX}:s_{,\Pi T}^*S:h_{,pX}S}_{(21)}+\underbrace{f_{,TT}(s_T^*)^{-1}s_{\Pi}^*:g_{,pX}S:h_{,pX}S}_{(22)}+\underbrace{f_{,T}(s_T^*)^{-1}s_{\Pi}^*:g_{,pX}S:h_{,pX}S}_{(23)}+ \quad (345)$$

$$\underbrace{+f_{,T}(s_T^*)^{-1}s_{,\Pi T}^*S:h_{,pX}S}_{(24)}\Big\}d\omega_X \quad (346)$$

and

$$\{\{\mathcal{G}, \mathcal{H}\}, \mathcal{F}\} \quad (347)$$

$$= \int_{\omega_X} \left\{ \underbrace{g_{,\chi\chi} \cdot h_{,p} \cdot f_{,p} - g_{,p} \cdot h_{,\chi\chi} \cdot f_{,p} - g_{,\chi} \cdot h_{,pp} \cdot f_{,\chi}}_{(2)} + \underbrace{g_{,pp} \cdot h_{,\chi} \cdot f_{,\chi}}_{(3)} + \right. \quad (348)$$

$$\underbrace{-g_{,pX}:h_{,\Pi\Pi}S:f_{,pX}S}_{(27)}-\underbrace{g_{,pX}:h_{,\Pi}S:f_{,pX}S}_{(28)}+\underbrace{g_{,\Pi\Pi}:h_{,pX}S:f_{,pX}S}_{(5)}+\underbrace{g_{,\Pi}:h_{,pX}S:f_{,pX}S}_{(6)}+ \quad (349)$$

$$\underbrace{+h_{,T\Pi}(s_T^*)^{-1}g_{,pX}:s_{\Pi}^*S:f_{,pX}S}_{(29)}+\underbrace{+h_{,T}(s_T^*)^{-1}g_{,pX}:s_{\Pi}^*S:f_{,pX}S}_{(30)}+\underbrace{+h_{,T}(s_T^*)^{-1}g_{,pX}:s_{,\Pi\Pi}^*S:f_{,pX}S}_{(31)}+ \quad (350)$$

$$\underbrace{+h_{,T}(s_T^*)^{-1}g_{,pX}:s_{,\Pi\Pi}^*S:f_{,pX}S}_{(32)}-\underbrace{g_{,T\Pi}(s_T^*)^{-1}s_{\Pi}^*:h_{,pX}S:f_{,pX}S}_{(17)}-\underbrace{g_{,T}(s_T^*)^{-1}s_{\Pi}^*:h_{,pX}S:f_{,pX}S}_{(10)}+ \quad (351)$$

$$\underbrace{-g_{,T}(s_T^*)^{-1}s_{,\Pi\Pi}^*S:h_{,pX}S:f_{,pX}S}_{(11)}-\underbrace{g_{,T}(s_T^*)^{-1}s_{\Pi}^*:h_{,pX}S:f_{,pX}S}_{(12)}+\underbrace{g_{,pX}:h_{,\Pi T}S(s_T^*)^{-1}s_{\Pi}^*:f_{,pX}S}_{(33)}+ \quad (352)$$

$$\underbrace{-g_{,\Pi T}:h_{,pX}S(s_T^*)^{-1}s_{\Pi}^*:f_{,pX}S}_{(9)}-\underbrace{h_{,TT}(s_T^*)^{-1}g_{,pX}:s_{\Pi}^*S:f_{,pX}S}_{(34)}-\underbrace{h_{,T}(s_T^*)^{-1}g_{,pX}:s_{,\Pi\Pi}^*S:f_{,pX}S}_{(35)}+ \quad (353)$$

$$\underbrace{-h_{,T}(s_T^*)^{-1}g_{,pX}:s_{,\Pi T}^*S:f_{,pX}S}_{(36)}+\underbrace{g_{,TT}(s_T^*)^{-1}s_{\Pi}^*:h_{,pX}S:f_{,pX}S}_{(19)}+\underbrace{g_{,T}(s_T^*)^{-1}s_{\Pi}^*:h_{,pX}S:f_{,pX}S}_{(20)}+ \quad (354)$$

$$\underbrace{+g_{,T}(s_T^*)^{-1}s_{,\Pi T}^*S:h_{,pX}S:f_{,pX}S}_{(21)}\Big\}d\omega_X \quad (355)$$

and, finally

$$\{\{\mathcal{H}, \mathcal{F}\}, \mathcal{G}\} \quad (356)$$

$$= \int_{\omega_X} \left\{ \underbrace{h_{,\chi\chi} \cdot f_{,p} \cdot g_{,p} - h_{,p} \cdot f_{,\chi\chi} \cdot g_{,p} - h_{,\chi} \cdot f_{,pp} \cdot g_{,\chi}}_{(25)} + \underbrace{h_{,pp} \cdot f_{,\chi} \cdot g_{,\chi}}_{(26)} + \right. \quad (357)$$

$$\underbrace{-h_{,pX}:f_{,\Pi\Pi}S:g_{,pX}S}_{(7)}-\underbrace{h_{,pX}:f_{,\Pi}S:g_{,pX}S}_{(8)}+\underbrace{h_{,\Pi\Pi}:f_{,pX}S:g_{,pX}S}_{(27)}+\underbrace{h_{,\Pi}:f_{,pX}S:g_{,pX}S}_{(28)}+ \quad (358)$$

$$\underbrace{+f_{,T\Pi}(s_T^*)^{-1}h_{,pX}:s_{\Pi}^*S:g_{,pX}S}_{(13)}+\underbrace{+f_{,T}(s_T^*)^{-1}h_{,pX}:s_{\Pi}^*S:g_{,pX}S}_{(14)}+\underbrace{+f_{,T}(s_T^*)^{-1}h_{,pX}:s_{,\Pi\Pi}^*S:g_{,pX}S}_{(15)}+ \quad (359)$$

$$\underbrace{+f_{,T}(s_T^*)^{-1}h_{,pX}:s_{,\Pi\Pi}^*S:g_{,pX}S}_{(16)}-\underbrace{h_{,T\Pi}(s_T^*)^{-1}s_{\Pi}^*:f_{,pX}S:g_{,pX}S}_{(33)}-\underbrace{h_{,T}(s_T^*)^{-1}s_{\Pi}^*:f_{,pX}S:g_{,pX}S}_{(30)}+ \quad (360)$$

$$\underbrace{-h_{,T}(s_T^*)^{-1}s_{,\Pi\Pi}^*S:f_{,pX}S:g_{,pX}S}_{(31)}-\underbrace{h_{,T}(s_T^*)^{-1}s_{\Pi}^*:f_{,pX}S:g_{,pX}S}_{(32)}+\underbrace{h_{,pX}:f_{,\Pi T}S(s_T^*)^{-1}s_{\Pi}^*:g_{,pX}S}_{(18)}+ \quad (361)$$

$$\underbrace{-h_{,\underline{\Pi}T} : f_{,pX\underline{\underline{S}}}(s_{,T}^*)^{-1} s_{,\underline{\Pi}}^* : g_{,pX\underline{\underline{S}}} - f_{,TT}(s_{,T}^*)^{-1} h_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* : g_{,pX\underline{\underline{S}}} - f_{,T}(s_{,T}^*)^{-1} h_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* : g_{,pX\underline{\underline{S}}} +}_{(29)} \underbrace{-f_{,T}(s_{,T}^*)^{-1} h_{,pX} : s_{,\underline{\Pi}T}^* : g_{,pX\underline{\underline{S}}} + h_{,TT}(s_{,T}^*)^{-1} s_{,\underline{\Pi}}^* : f_{,pX\underline{\underline{S}}} : g_{,pX\underline{\underline{S}}} + h_{,T}(s_{,T}^*)^{-1} s_{,\underline{\Pi}}^* : f_{,pX\underline{\underline{S}}} : g_{,pX\underline{\underline{S}}} +}_{(22)} \underbrace{-f_{,T}(s_{,T}^*)^{-1} h_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* : g_{,pX\underline{\underline{S}}} + h_{,T}(s_{,T}^*)^{-1} s_{,\underline{\Pi}}^* : f_{,pX\underline{\underline{S}}} : g_{,pX\underline{\underline{S}}} +}_{(23)} \quad (362)$$

$$\underbrace{-f_{,T}(s_{,T}^*)^{-1} h_{,pX} : s_{,\underline{\Pi}T}^* : g_{,pX\underline{\underline{S}}} + h_{,TT}(s_{,T}^*)^{-1} s_{,\underline{\Pi}}^* : f_{,pX\underline{\underline{S}}} : g_{,pX\underline{\underline{S}}} + h_{,T}(s_{,T}^*)^{-1} s_{,\underline{\Pi}}^* : f_{,pX\underline{\underline{S}}} : g_{,pX\underline{\underline{S}}} +}_{(24)} \underbrace{-f_{,T}(s_{,T}^*)^{-1} h_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* : g_{,pX\underline{\underline{S}}} + h_{,T}(s_{,T}^*)^{-1} s_{,\underline{\Pi}}^* : f_{,pX\underline{\underline{S}}} : g_{,pX\underline{\underline{S}}} +}_{(35)} \underbrace{-f_{,T}(s_{,T}^*)^{-1} h_{,pX} : s_{,\underline{\Pi}\underline{\underline{S}}}^* : g_{,pX\underline{\underline{S}}} + h_{,T}(s_{,T}^*)^{-1} s_{,\underline{\Pi}}^* : f_{,pX\underline{\underline{S}}} : g_{,pX\underline{\underline{S}}} +}_{(36)} \quad (363)$$

$$\underbrace{+h_{,T}(s_{,T}^*)^{-1} s_{,\underline{\Pi}T}^* : f_{,pX\underline{\underline{S}}} : g_{,pX\underline{\underline{S}}} +}_{(37)} d\omega_X \quad (364)$$

The identification of the terms one by one in the previous definitions show that

$$\{\{\mathcal{F}, \mathcal{G}\}, \mathcal{H}\} + \{\{\mathcal{G}, \mathcal{H}\}, \mathcal{F}\} + \{\{\mathcal{H}, \mathcal{F}\}, \mathcal{G}\} = 0$$

for all functinonal $\mathcal{F}, \mathcal{G}, \mathcal{H}$ such that their respective densities f, g, h meet the conditions (321) to (326), and for a second Piola Kirchhoff stress tensor fulfilling conditions listed in (327).

qed. ■

2.9. Linearization of the Sidoroff decomposition

Proposition 2.8

The decomposition of the gradient of transformation into an elastic and an irreversible part

$$\underline{\underline{F}} = \underline{\underline{F}}_e \cdot \underline{\underline{F}}_i \quad (365)$$

leads to

$$\underline{\underline{\varepsilon}} = \underline{\underline{\varepsilon}}_e + \underline{\underline{\varepsilon}}_i \quad (366)$$

provided that

$$\left\| \frac{\partial \underline{\underline{u}}_e}{\partial \underline{\underline{X}}} \right\| \ll 1, \quad \left\| \frac{\partial \underline{\underline{u}}_i}{\partial \underline{\underline{X}}} \right\| \ll 1 \quad (367)$$

Proof _____

$$\underline{\underline{F}} = \frac{\partial \underline{\underline{u}}}{\partial \underline{\underline{X}}} + \underline{\underline{I}} \Rightarrow \frac{1}{2} (\underline{\underline{F}} + \underline{\underline{F}}^\top) = \underline{\underline{\varepsilon}} + \underline{\underline{I}} \quad (368)$$

$$\Rightarrow \underline{\underline{\varepsilon}} = -\underline{\underline{I}} + \frac{1}{2} (\underline{\underline{F}}_e \cdot \underline{\underline{F}}_i + \underline{\underline{F}}_i^\top \cdot \underline{\underline{F}}_e^\top) \quad (369)$$

$$\Rightarrow \underline{\underline{\varepsilon}} = \underbrace{\frac{1}{2} \left(\frac{\partial \underline{\underline{u}}_e}{\partial \underline{\underline{X}}} + \frac{\partial \underline{\underline{u}}_e^\top}{\partial \underline{\underline{X}}} \right)}_{\underline{\underline{\varepsilon}}_e} + \underbrace{\frac{1}{2} \left(\frac{\partial \underline{\underline{u}}_i}{\partial \underline{\underline{X}}} + \frac{\partial \underline{\underline{u}}_i^\top}{\partial \underline{\underline{X}}} \right)}_{\underline{\underline{\varepsilon}}_i} + \frac{1}{2} \left(\frac{\partial \underline{\underline{u}}_e}{\partial \underline{\underline{X}}} \cdot \frac{\partial \underline{\underline{u}}_i}{\partial \underline{\underline{X}}} + \frac{\partial \underline{\underline{u}}_i^\top}{\partial \underline{\underline{X}}} \cdot \frac{\partial \underline{\underline{u}}_e^\top}{\partial \underline{\underline{X}}} \right) \quad (370)$$

Consider that

$$\left\| \frac{\partial \underline{\underline{u}}_e}{\partial \underline{\underline{X}}} \right\| \ll 1, \quad \left\| \frac{\partial \underline{\underline{u}}_i}{\partial \underline{\underline{X}}} \right\| \ll 1 \quad (371)$$

so that we can neglect the second order terms in the expression of $\underline{\underline{\varepsilon}}$, such that

$$\underline{\underline{\varepsilon}} = \frac{1}{2} \left(\frac{\partial \underline{\underline{u}}_e}{\partial \underline{\underline{X}}} + \frac{\partial \underline{\underline{u}}_e^\top}{\partial \underline{\underline{X}}} \right) + \frac{1}{2} \left(\frac{\partial \underline{\underline{u}}_i}{\partial \underline{\underline{X}}} + \frac{\partial \underline{\underline{u}}_i^\top}{\partial \underline{\underline{X}}} \right) = \underline{\underline{\varepsilon}}_e + \underline{\underline{\varepsilon}}_i \quad (372)$$

qed. ■

References

- Bergstrom, J (May 1998). “Constitutive Modeling of the Large Strain Time-Dependent Behavior of Elastomers”. In: *Journal of the Mechanics and Physics of Solids* 46.5, pp. 931–954. ISSN: 00225096. DOI: 10.1016/S0022-5096(97)00075-6.
- Coleman, Bernard D. and Morton E. Gurtin (July 1967). “Thermodynamics with Internal State Variables”. In: *The Journal of Chemical Physics* 47.2, pp. 597–613. ISSN: 0021-9606, 1089-7690. DOI: 10.1063/1.1711937.
- Itskov, Mikhail (2019). *Tensor Algebra and Tensor Analysis for Engineers: With Applications to Continuum Mechanics*. Mathematical Engineering. Cham: Springer International Publishing. ISBN: 978-3-319-98805-4 978-3-319-98806-1. DOI: 10.1007/978-3-319-98806-1.
- Kamasamudram, Vasudevan, Michel Coret, and Nicolas Moës (June 2023). “Computation of Energy Evolution during the Dynamic Fracture of Elastomers Using the Finite Viscoelastic Model: Its Implementation in Abaqus”. In: *Mechanics of Materials* 181, p. 104638. ISSN: 01676636. DOI: 10.1016/j.mechmat.2023.104638.
- Lemaitre, Jean and Jean-Louis Chaboche (2001). *Mécanique des matériaux solides*. 2e éd. Sciences sup. Paris: Dunod. ISBN: 978-2-10-005662-0.
- Reese, Stefanie and Sanjay Govindjee (1997). “Theoretical and Numerical Aspects in the Thermo-Viscoelastic Material Behaviour of Rubber-like Polymers”. In: *Mechanics of Time-Dependent Materials* 1.4, pp. 357–396. ISSN: 13852000. DOI: 10.1023/A:1009795431265.
- (Sept. 1998). “A Theory of Finite Viscoelasticity and Numerical Aspects”. In: *International Journal of Solids and Structures* 35.26-27, pp. 3455–3482. ISSN: 00207683. DOI: 10.1016/S0020-7683(97)00217-5.
- Schiebl, Mark and Peter Betsch (July 2021). “Structure-preserving Space-time Discretization of Large-strain Thermo-viscoelasticity in the Framework of GENERIC”. In: *International Journal for Numerical Methods in Engineering* 122.14, pp. 3448–3488. ISSN: 0029-5981, 1097-0207. DOI: 10.1002/nme.6670.